PHOT 451: Microscale optical system design
LECTURE 03

Michaël Barbier, Fall semester (2026-2027)

OVERVIEW OF THE COURSE

week	Topic
Week 1	Introduction to micro-scale optical components
Week 2	Light propagation in free space
Week 3	Geometric optics and raycasting
Week 4	Diffraction limit & Abberations
Week 5	Beam propagation
Week 6	Refractive optical elements Microlenses
Week 7	**Midterm** + Blazed Fresnel lenses
Week 8	Digital lenses
Week 9	Diffractive optical elements
Week 10	Wave guides and beam propagation
Week 11	Wave mixing
Week 12	**Project presentations**
Week 13	photonic crystals, Gratings, periodic structures
Week 14	Whole optical system optimization

Geometrical Optics (Ray Optics)

GEOMETRIC OPTICS

- Light path: straight line in homogeneous medium
- Refractive index:

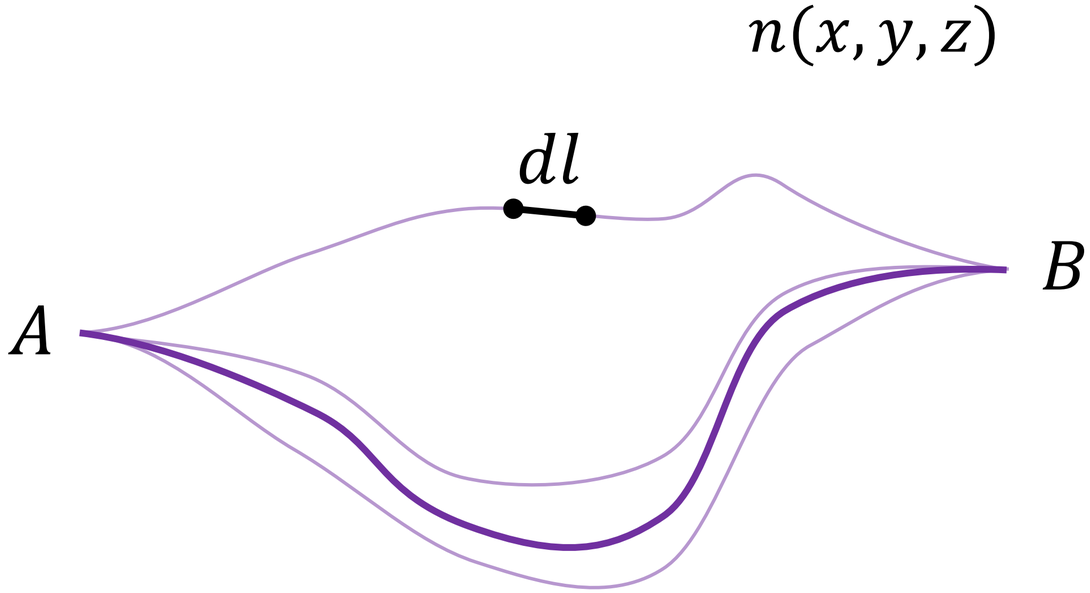
$$n = \frac{c}{v} = \frac{\text{velocity vacuum}}{\text{(phase) velocity medium}}$$

- Optical path length:

$$\text{OPL} = \int_A^B n(x,y,z)\, dl$$

- Fermat's principle:

$$\delta\left[\int_A^B n(x,y,z)\, dl\right] = 0$$



GEOMETRIC OPTICS & RAY-TRACING

- Phase local wavefront (eikonal)

$$\varphi(\boldsymbol{r}) = \boldsymbol{k} \cdot \boldsymbol{r}$$

- Wavefront: line constant phase

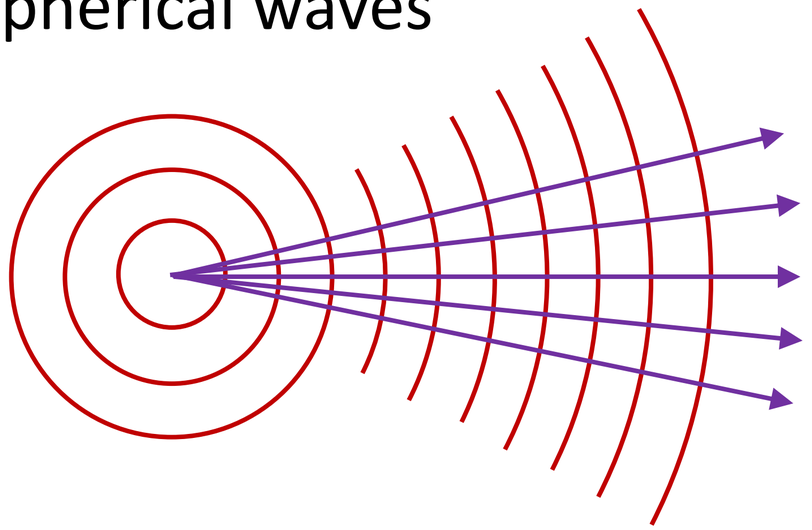
- Light rays perpendicular to wave fronts (requires isotropic medium)

Parallel rays ⟺ Plane waves

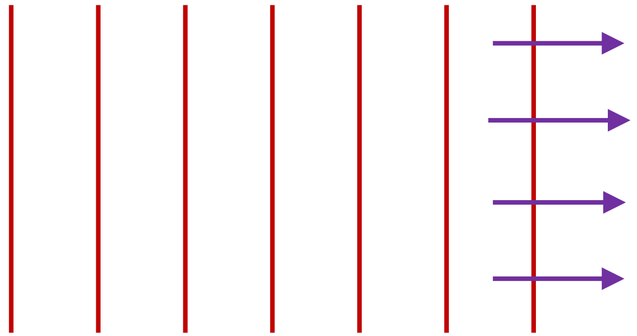
Point source ⟺ Spherical waves

Rays follow local k vector

Spherical waves



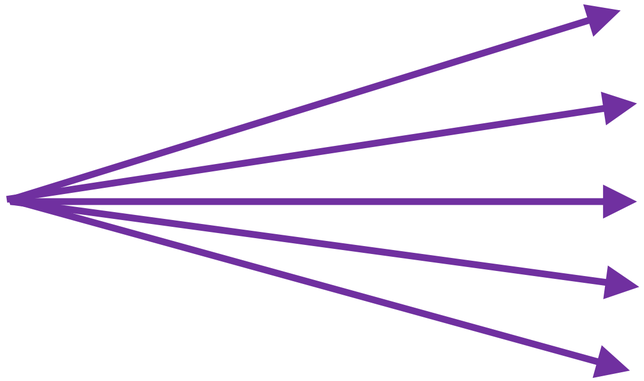
Plane waves



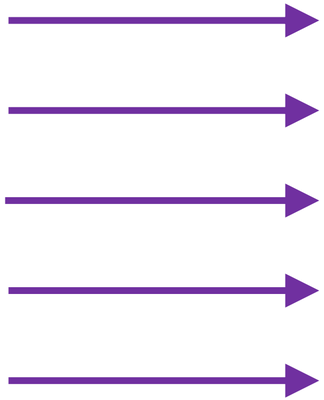
RAY BUNDLES OF LIGHT

- Rays go **straight** in **homogeneous medium**
- **Ray bundles** represent an angular portion of spherical waves (diverging or converging)
- Ray bundles of interest:

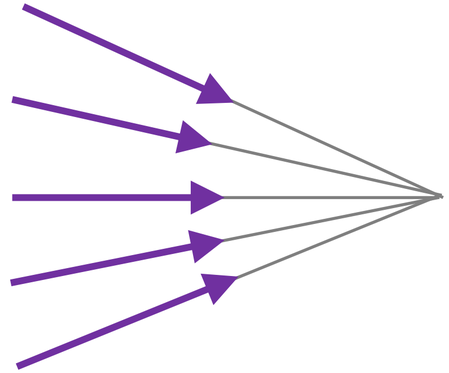
Rays from a **point source**



Parallel rays (**Collimation**)

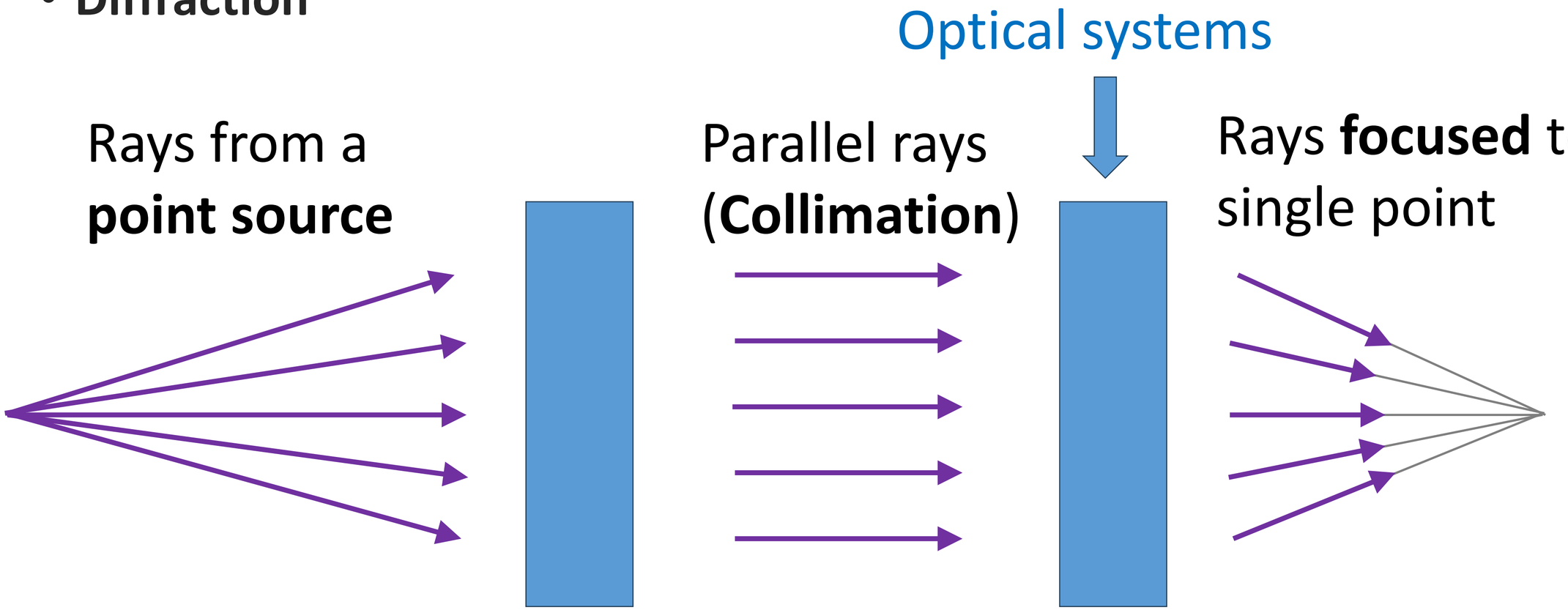


Rays **focused** to single point



COLLIMATION & FOCUSING

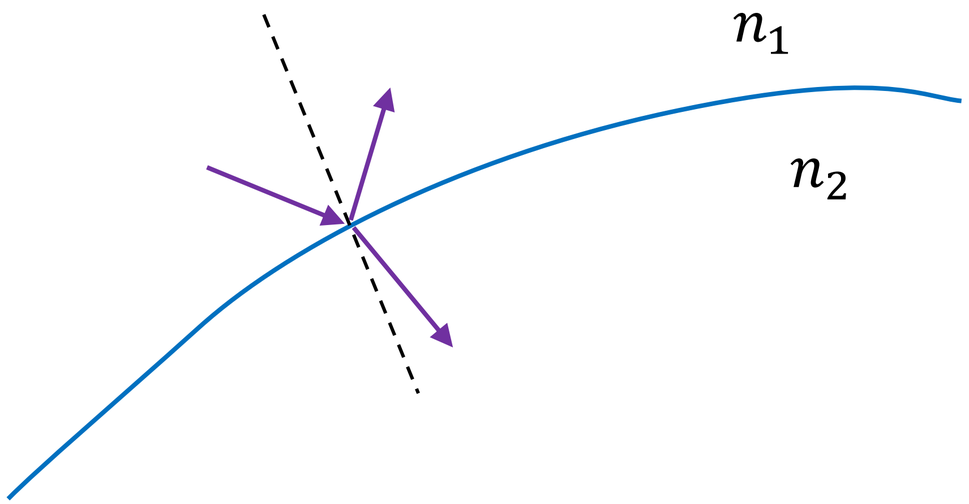
- Changing the direction of light rays
  - **Heterogeneous medium**: Gradient Index **(GRIND)**
  - **Refraction, reflection**
  - **Diffraction**



REFRACTION OF RAYS AT SURFACE

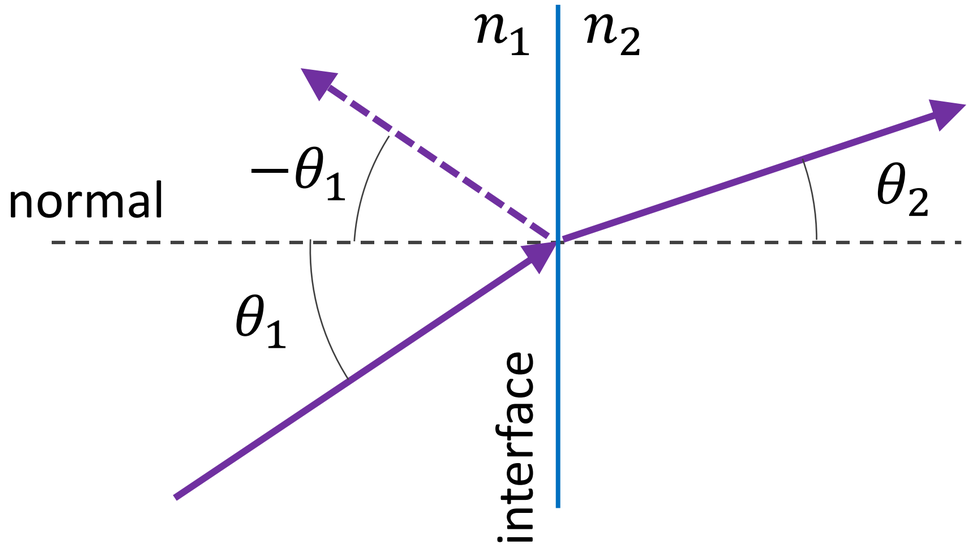
- Wavelength ≪ curvature radius

$$\lambda \ll R_c$$



- **Refraction: Snell's law**

$$n_1 \sin(\theta_1) = \mathrm{n}_2 \sin(\theta_2)$$



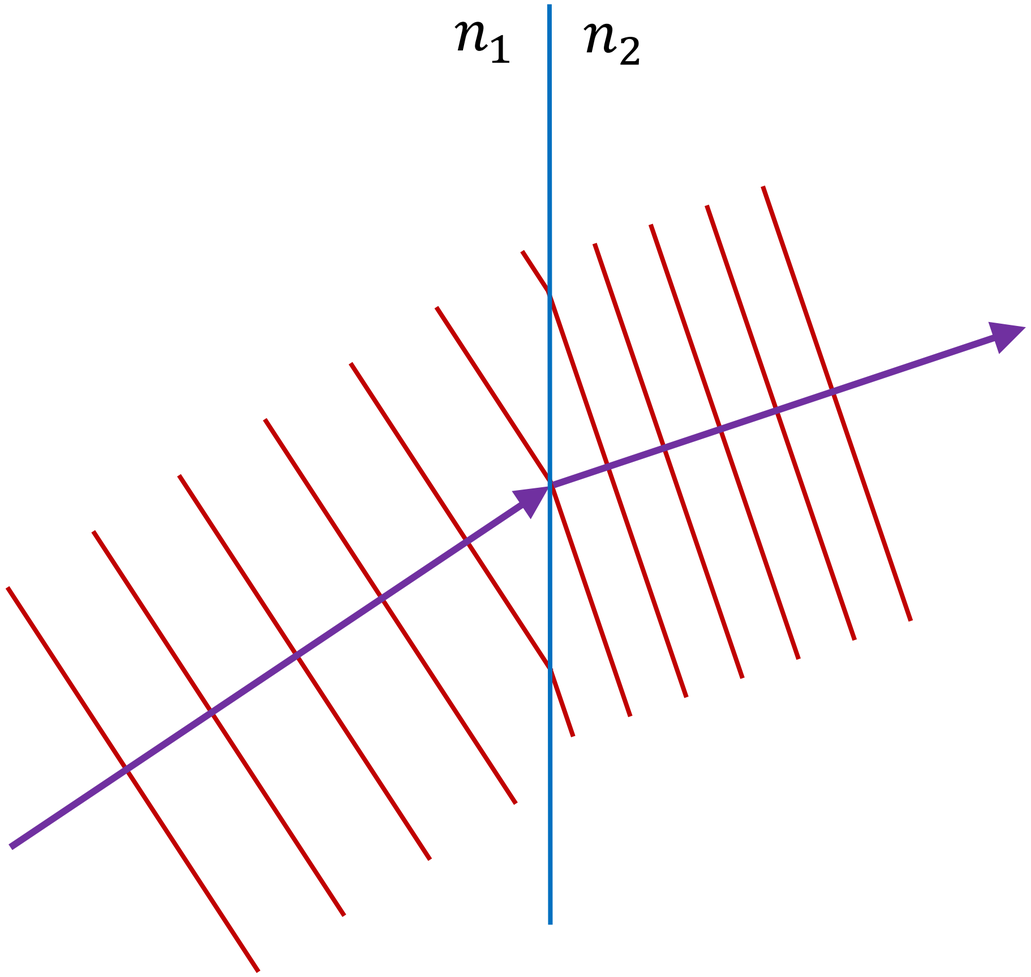
FERMAT'S PRINCIPLE & WAVE FRONTS

- Phase along rays:

$$\varphi(\boldsymbol{r}) = \boldsymbol{k} \cdot \boldsymbol{r}$$

- Matching wave fronts

$$n_j = \frac{c_0}{v_j} = \frac{\text{velocity vacuum}}{\text{velocity medium}}$$



FERMAT'S PRINCIPLE & WAVE FRONTS

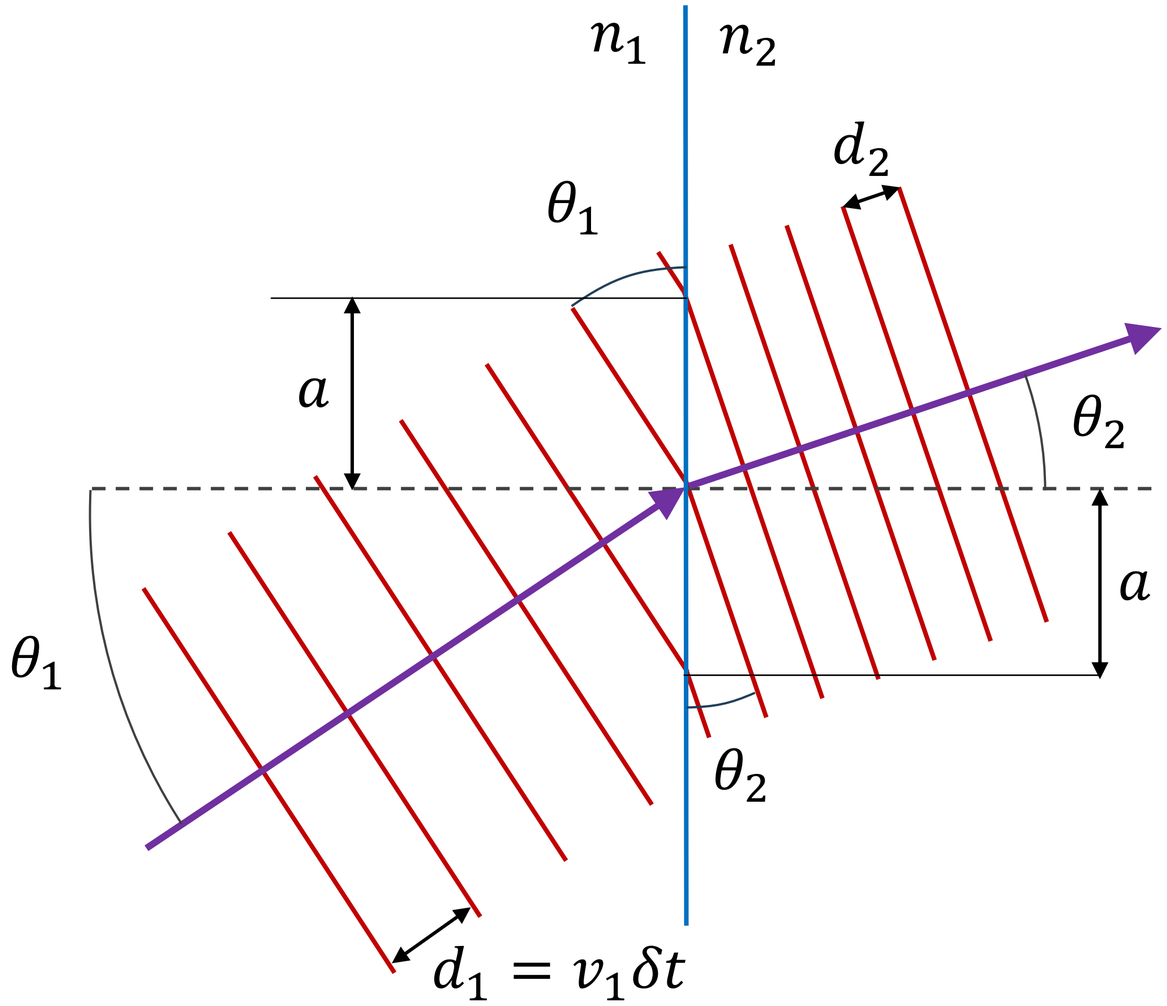
- Phase along rays:

$$\varphi(\boldsymbol{r}) = \boldsymbol{k} \cdot \boldsymbol{r}$$

- Matching wave fronts

$$n_j = \frac{c_0}{v_j} = \frac{\text{velocity vacuum}}{\text{velocity medium}}$$

$$\begin{cases} a = \dfrac{d_1}{\sin\theta_1} = \dfrac{v_1 \delta t}{\sin\theta_1} = \dfrac{c_0 \delta t}{n_1 \sin\theta_1} \\ a = \dfrac{d_2}{\sin\theta_2} = \dfrac{v_2 \delta t}{\sin\theta_2} = \dfrac{c_0 \delta t}{n_2 \sin\theta_2} \end{cases}$$



FERMAT'S PRINCIPLE & WAVE FRONTS

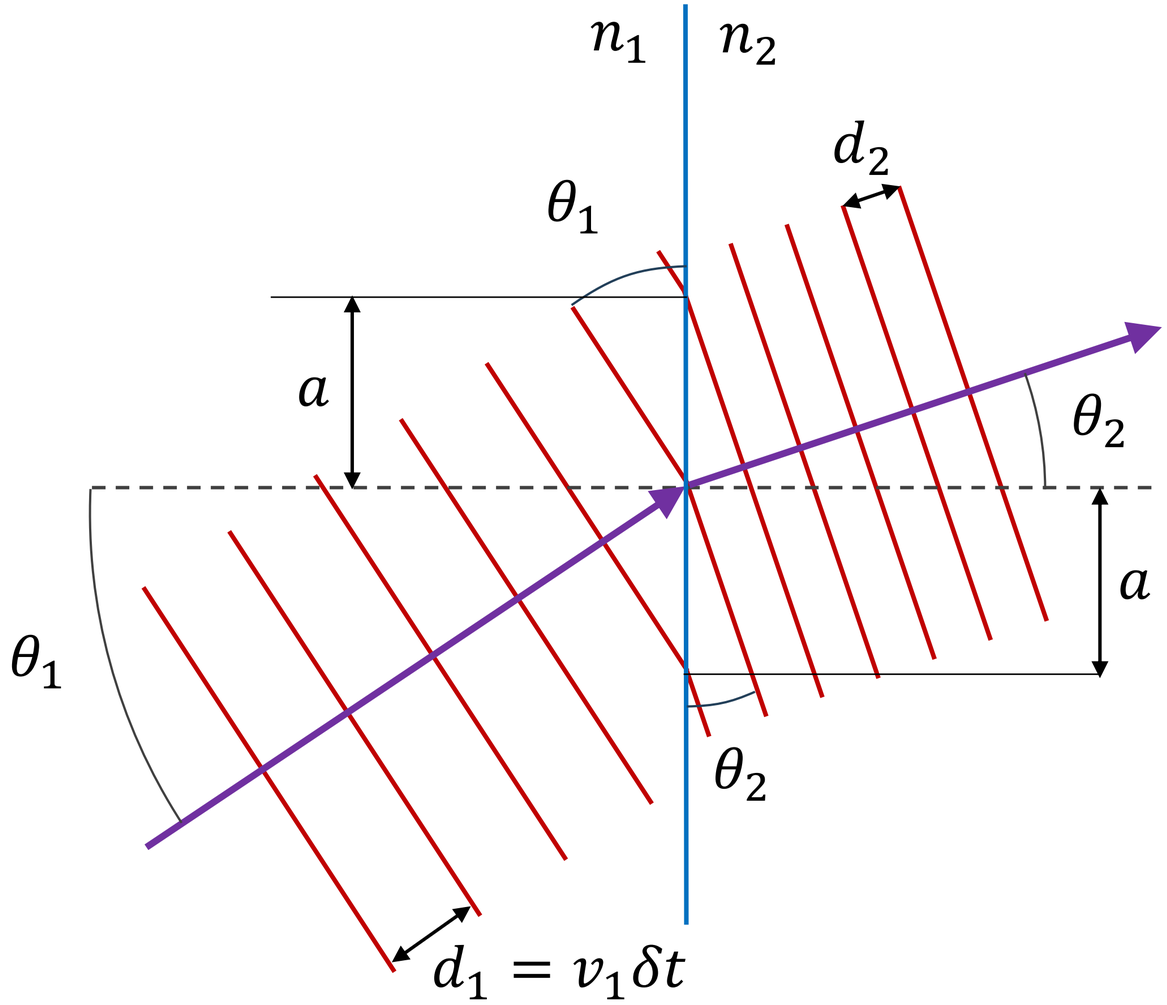
- Phase along rays:

$$\varphi(\boldsymbol{r}) = \boldsymbol{k} \cdot \boldsymbol{r}$$

- Matching wave fronts

$$\Rightarrow \quad \frac{c_0 \delta t}{n_1 \sin\theta_1} = \frac{c_0 \delta t}{n_2 \sin\theta_2}$$

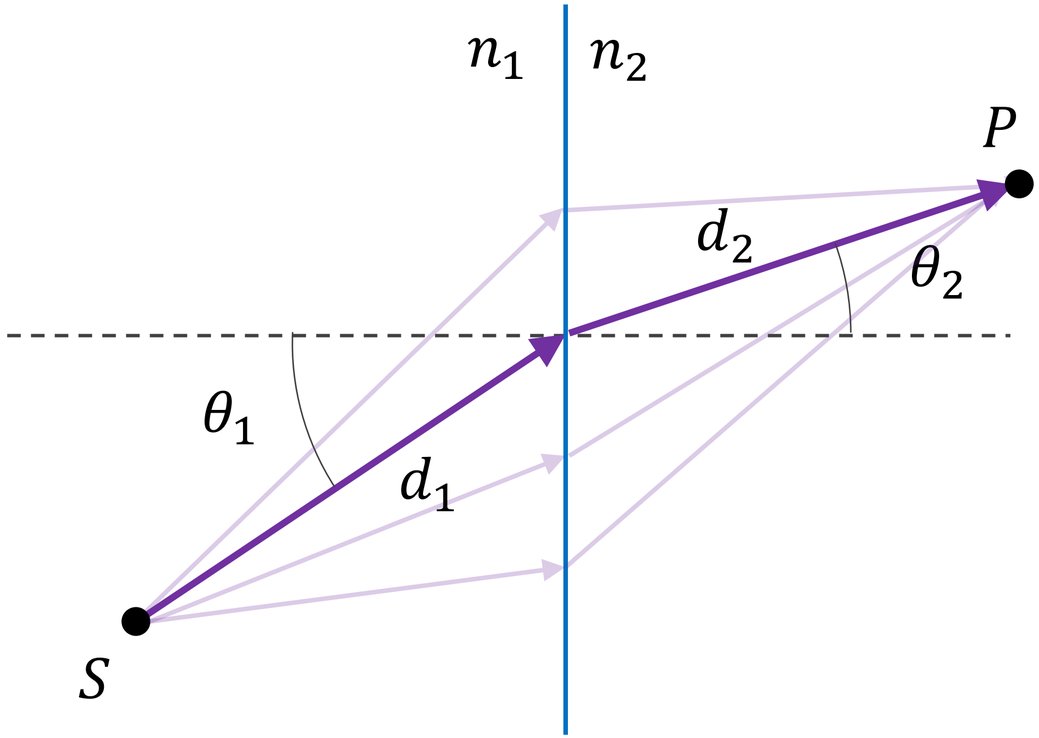
$$\Rightarrow \quad n_1 \sin(\theta_1) = \mathrm{n}_2 \sin(\theta_2)$$



FERMAT'S PRINCIPLE & WAVE FRONTS

- **Fermat's** principle

  Light takes path of least time



Minimize time Δt :

$$\Delta t = \Delta t_1 + \Delta t_2 = \frac{d_1}{v_1} + \frac{d_2}{v_2}$$

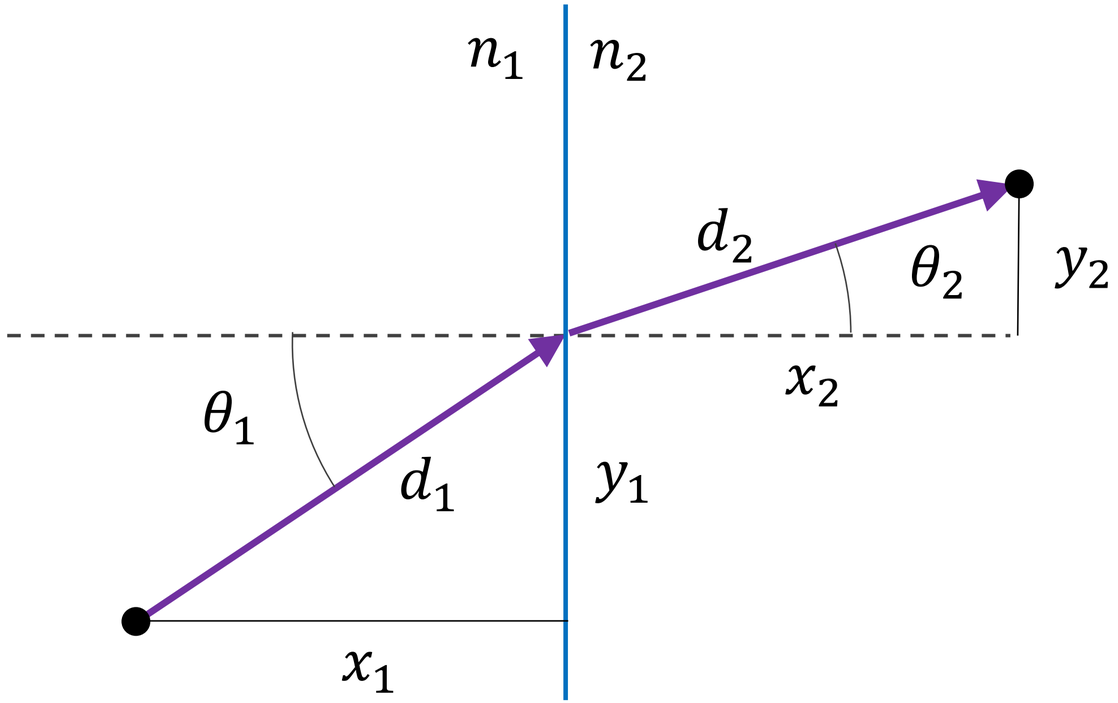
Condition:

$$\frac{\partial\, \Delta t}{\partial z_1} = 0$$

FERMAT'S PRINCIPLE & WAVE FRONTS

- **Fermat's** principle

  Light takes path of least time



Minimize time Δt :

$$\Delta t = \Delta t_1 + \Delta t_2 = \frac{d_1}{v_1} + \frac{d_2}{v_2}$$

$$= \frac{\sqrt{x_1^2 + y_1^2}}{v_1} + \frac{\sqrt{x_2^2 + (y - y_1)^2}}{v_2}$$

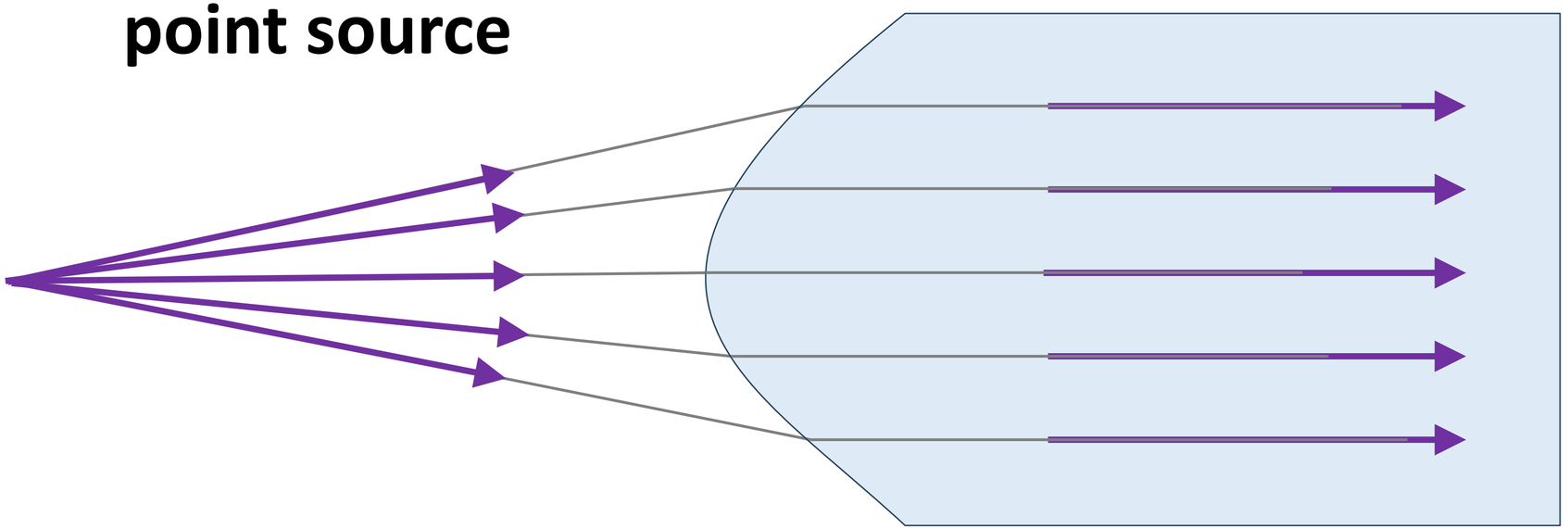
$$\frac{\partial\, \Delta t}{\partial y_1} = 0 \quad \Rightarrow \quad \frac{2y_1}{v_1\sqrt{x_1^2 + y_1^2}} = \frac{2(y - y_1)}{v_2\sqrt{x_2^2 + (y - y_1)^2}} \quad \Rightarrow \quad n_1 \sin(\theta_1) = \mathrm{n}_2 \sin(\theta_2)$$

REFRACTION: STIGMATIC SURFACES

- Changing the direction of light rays
- Perfect surface? Conic surfaces
- Example: Diverging rays (point source) to collimation

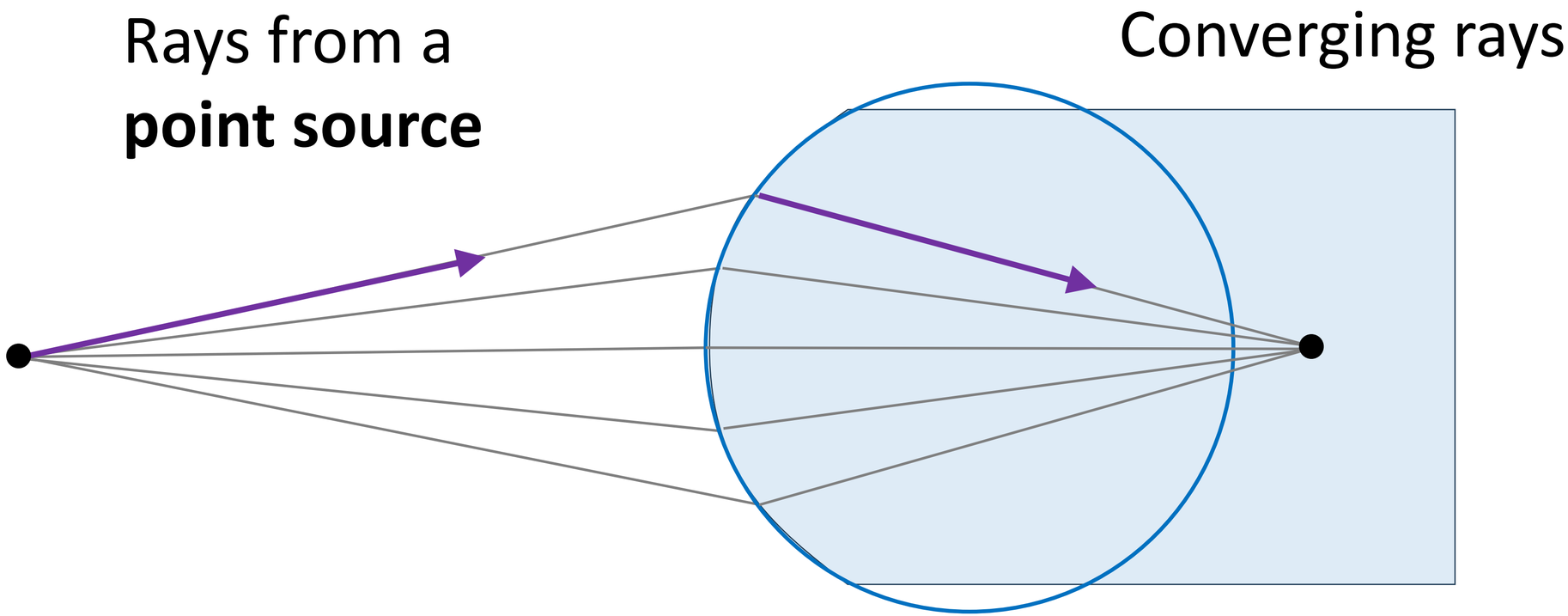
Rays from a
point source

Parallel rays
(**Collimation**)



REFRACTION: SPHERICAL SURFACES

- Not perfect
- Easier to produce
- Good performance for paraxial rays

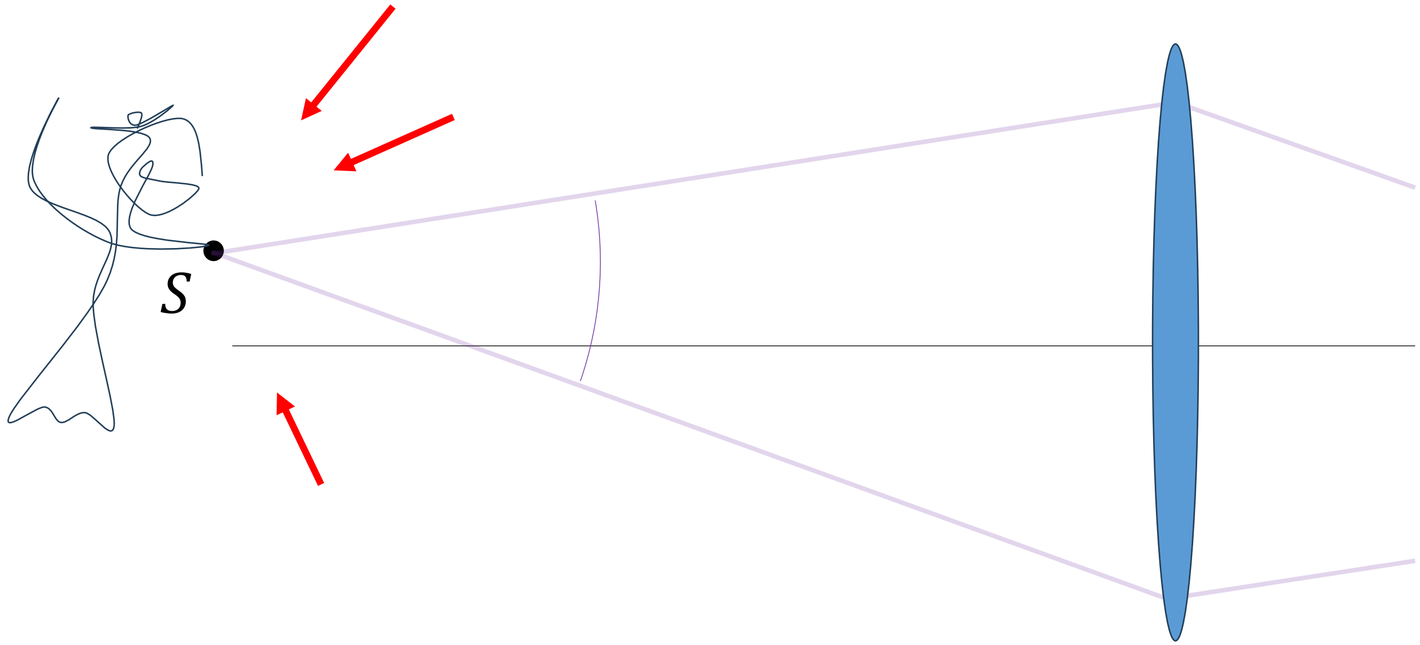


Optical Systems for imaging

IMAGING

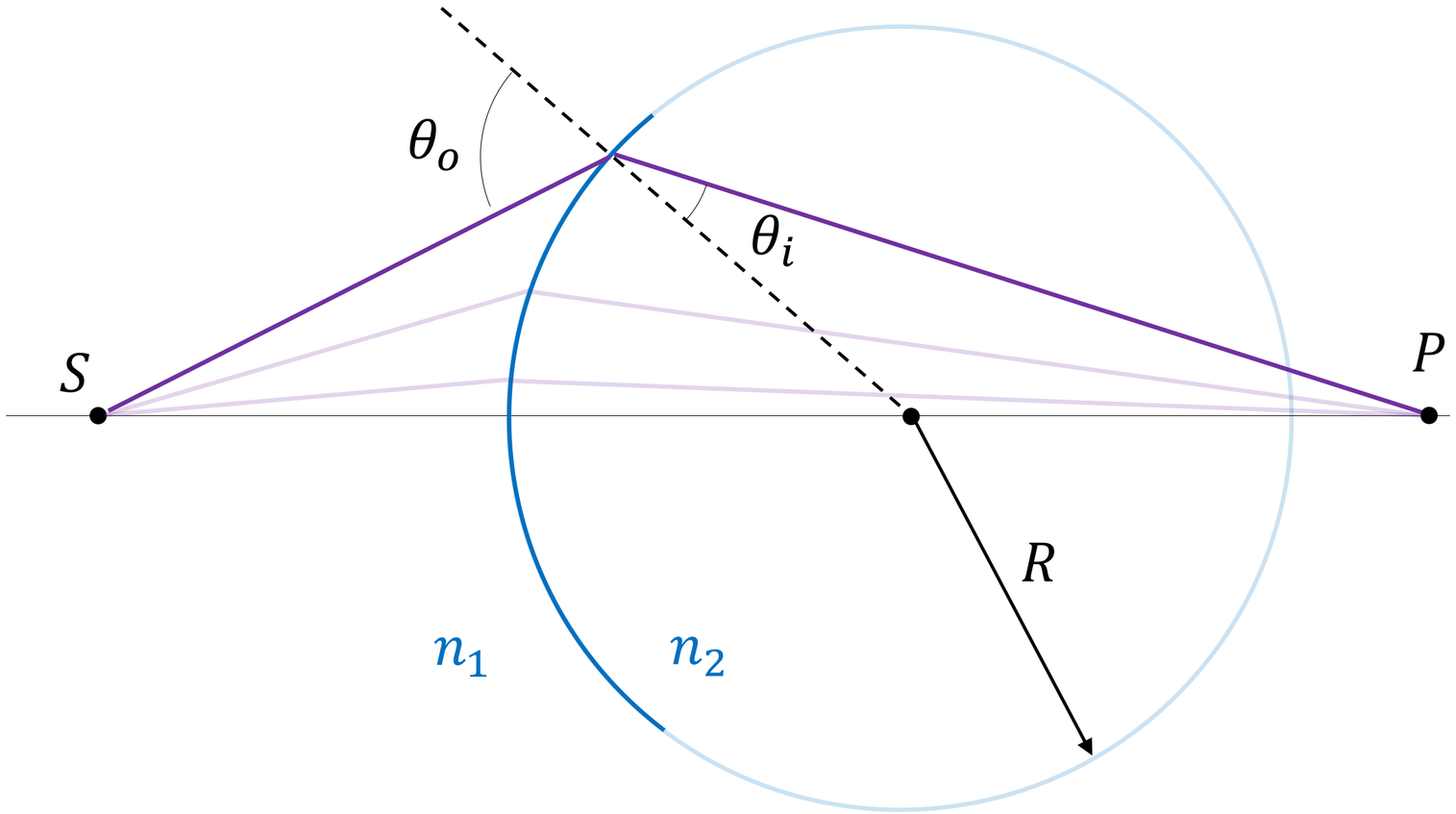
- Light reflected from object

- Small angle of light captured by optical system



DIRECTING LIGHT: SPHERICAL SURFACES

- Single surface
  - Curvature radius R
  - Refraction index n_2
- At surface: Snell's law
- Rays from S to P ?
- Apply Fermat's principle

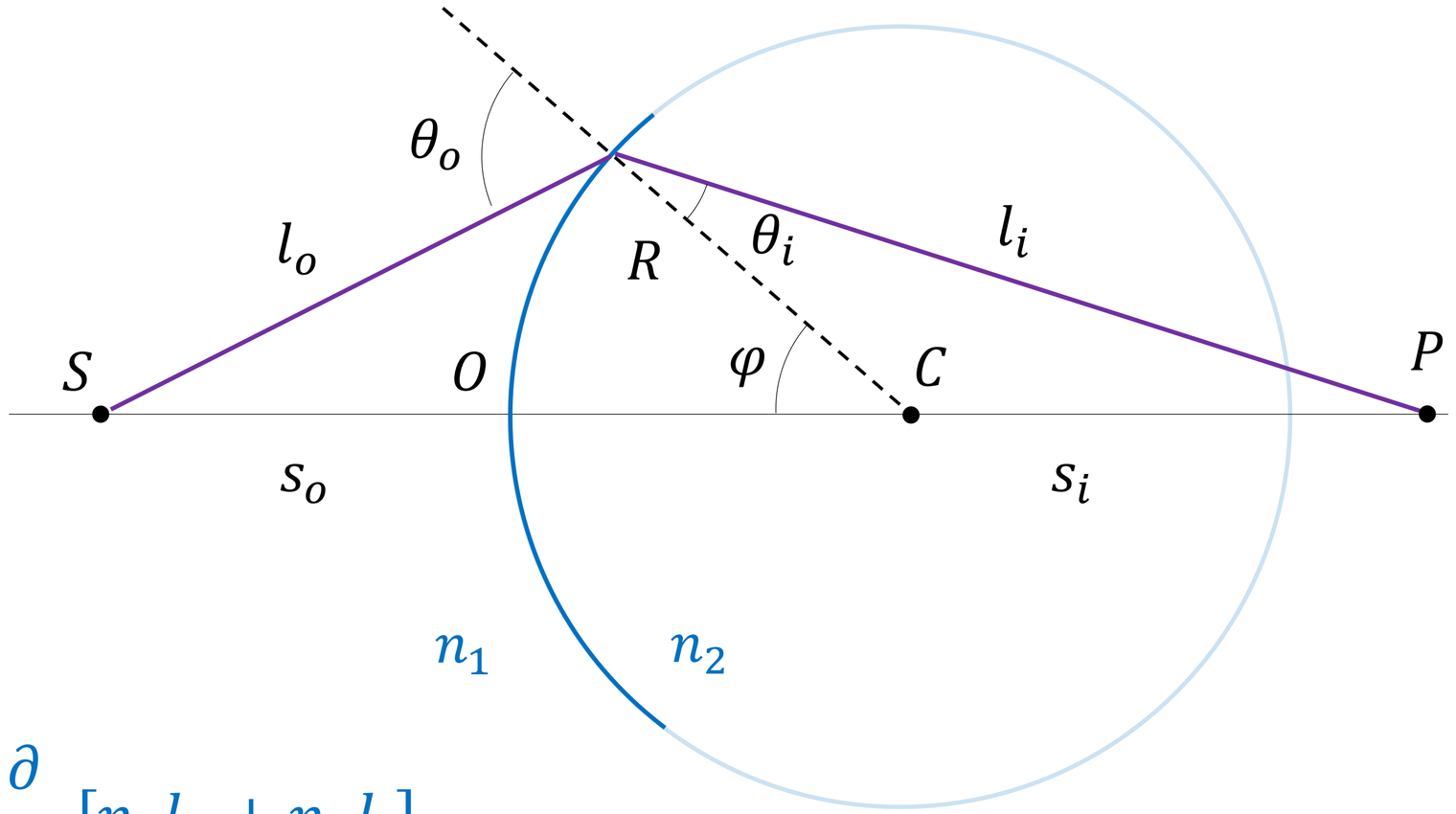


SPHERICAL SURFACES

- Optical Path Length

$$\mathrm{OPL} = \int_A^B n(x,y,z)\,dl$$

$$= n_1 l_o + n_2 l_i$$



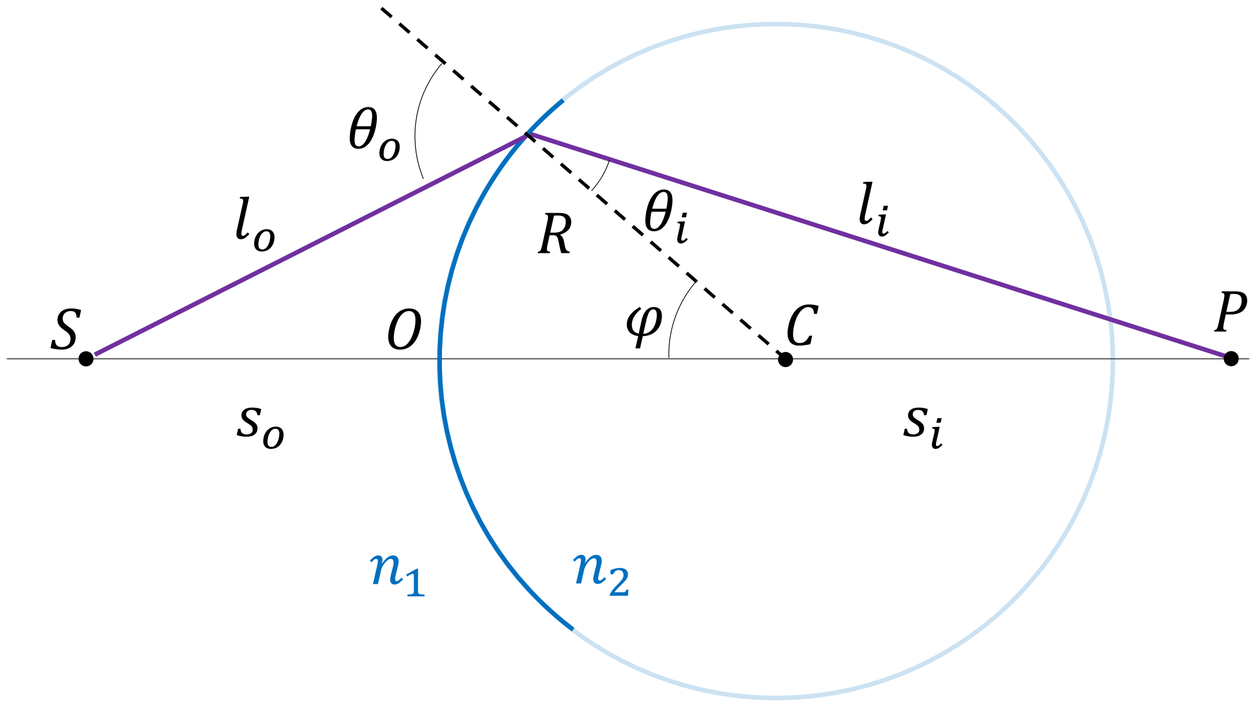
- Fermat's principle

$$0 = \delta\left[\int_A^B n(x,y,z)\,dl\right] = \frac{\partial}{\partial\varphi}\left[n_1 l_o + n_2 l_i\right]$$

$$\Rightarrow \quad \frac{1}{R}\left[-\frac{n_1 s_o}{l_o} + \frac{n_2 s_i}{l_i}\right] = \frac{n_1}{l_o} + \frac{n_2}{l_i}$$

SPHERICAL SURFACES

$$\frac{1}{R}\left[-\frac{n_1 s_o}{l_o}+\frac{n_2 s_i}{l_i}\right]=\frac{n_1}{l_o}+\frac{n_2}{l_i}$$



- Problem: depends on φ

- Paraxial approximation

$\cos\varphi \approx 1 \;\;\&\;\; s_o \approx l_o \;\;\&\;\; s_i \approx l_i$

$$\frac{n_2-n_1}{R}=\frac{n_1}{s_o}+\frac{n_2}{s_i}$$

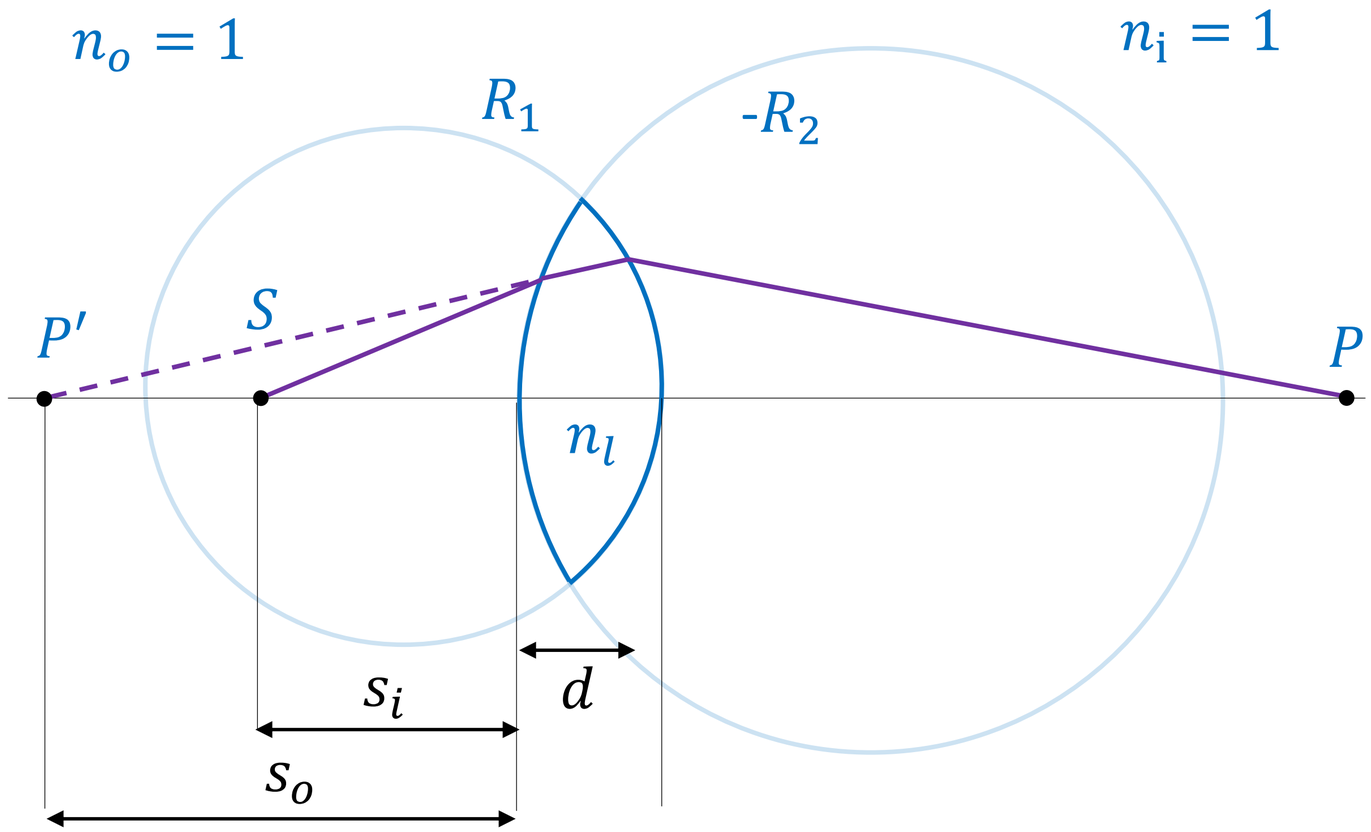
LENSMAKER'S FORMULA

- Paraxial approximation

- Thin lenses, in air

$$n_o = n_i = 1$$

- Perfect focus, image in plane



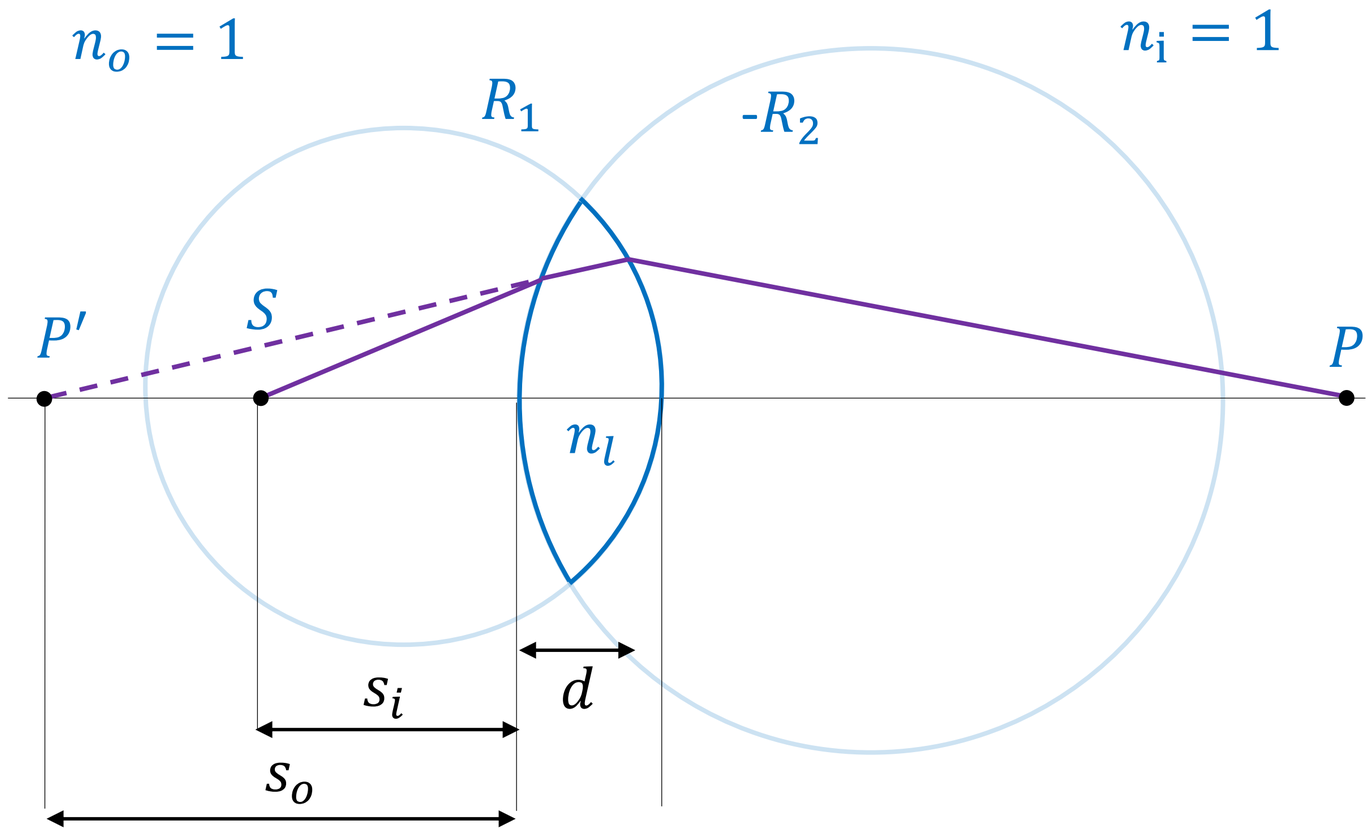
A THIN LENS

- Paraxial approximation

- Thin lenses, in air

$$n_o = n_i = 1$$

- Perfect focus, image in plane

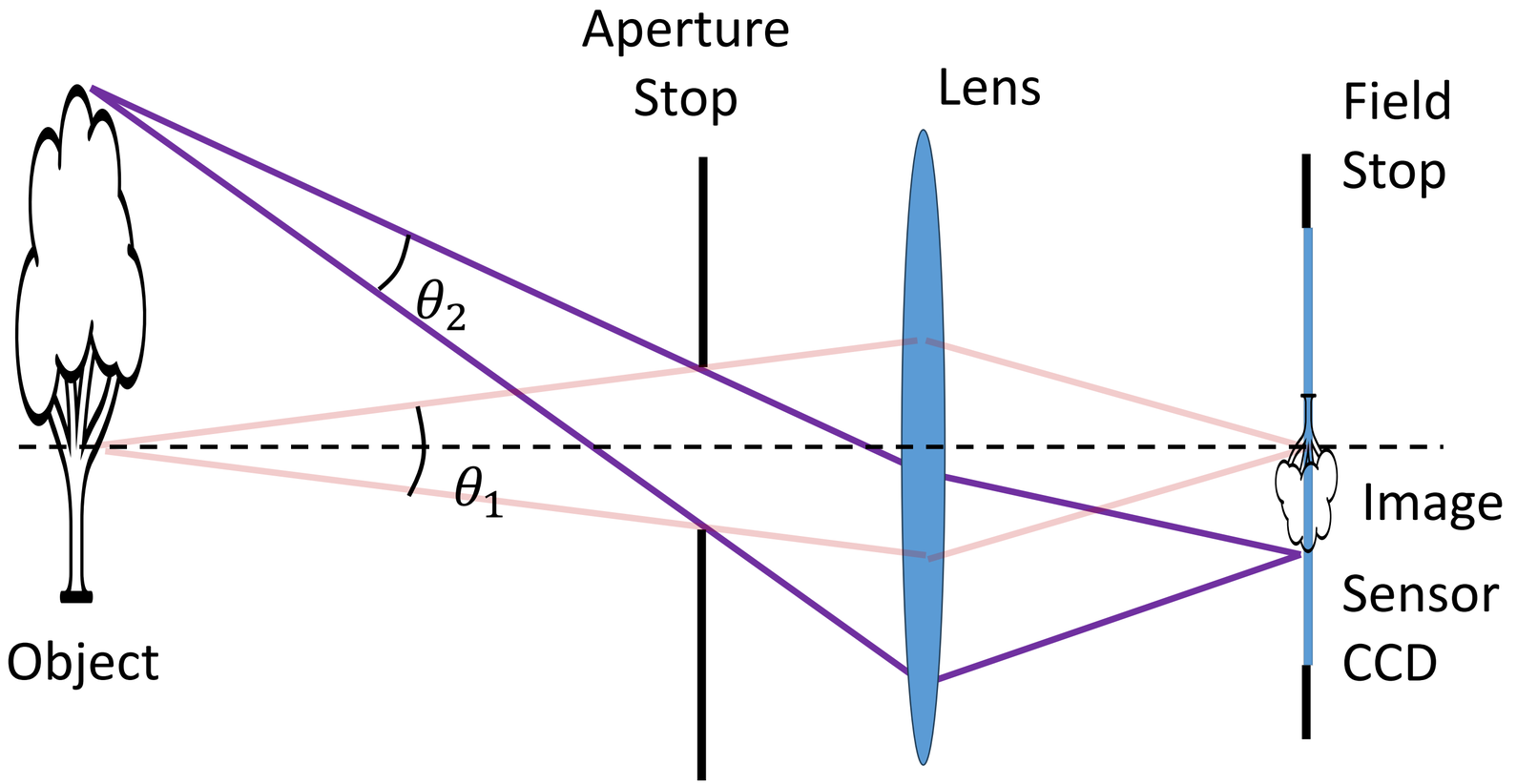


- The Lensmaker's formula

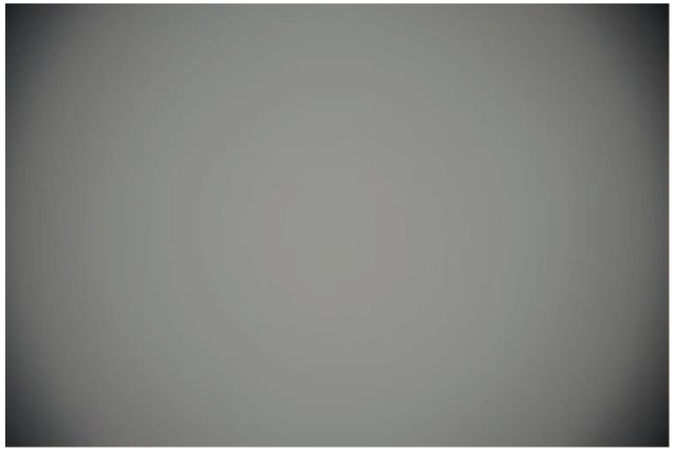
$$\frac{1}{s_\mathrm{o}} + \frac{1}{s_\mathrm{i}} = \frac{1}{f} = (n_l - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

WHOLE IMAGING SYSTEM

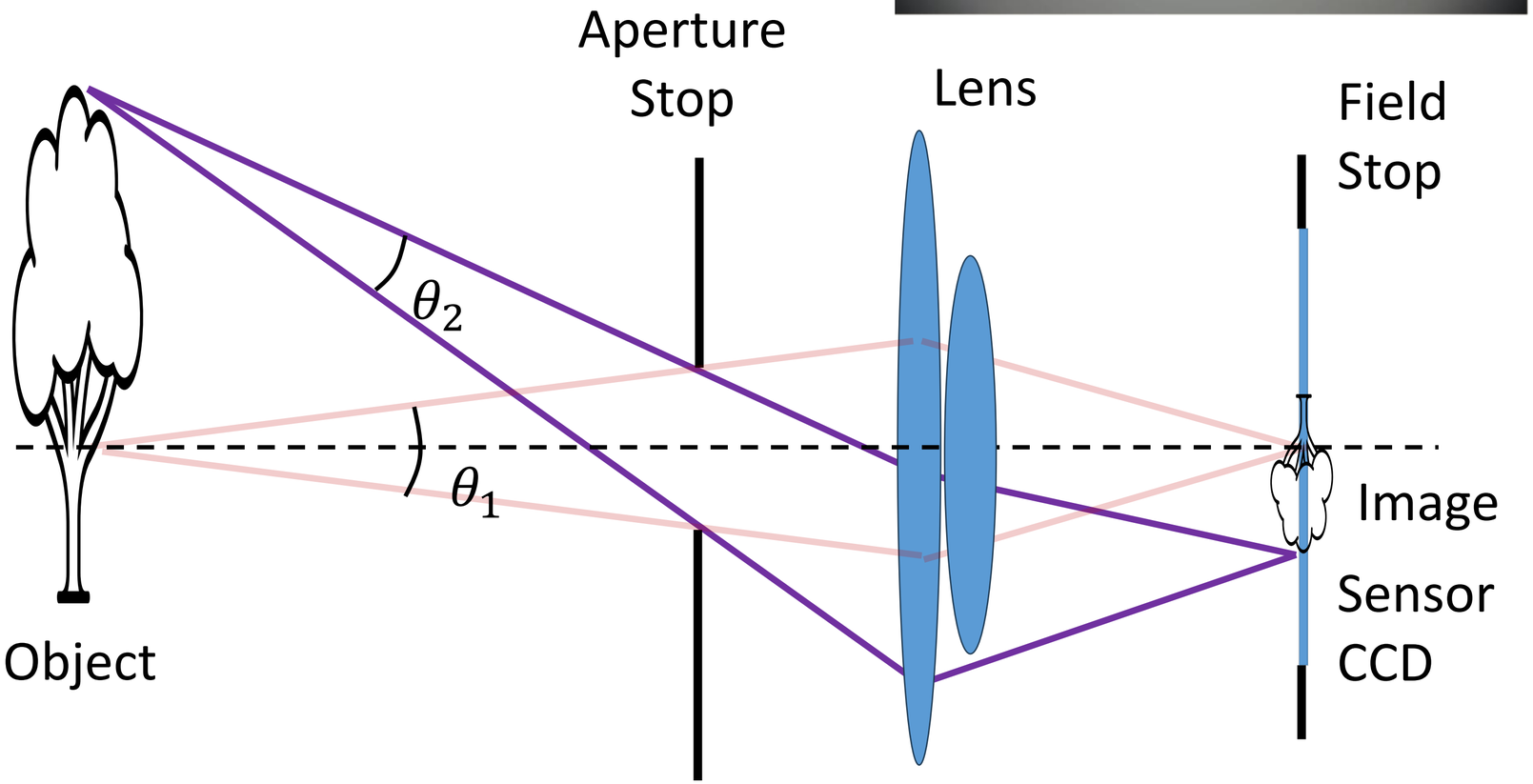
- Lenses
- Aperture stops
- Sensor, sensor size and field stops



VIGNETTING



- Intensity depends on angle θ
- Angle $\theta_2 < \theta_1$
- Edges have less intensity
- Apertures and clear lens diameters are important



Methods of Light Propagation

Some insight in different models

SUMMARY LIGHT PROPAGATION

- Full wave electromagnetic field
- Wave equation: Amplitude and Phase
  - Near field effects
- Paraxial approximation
  - Geometrical Optics: spherical lenses
  - Paraxial Helmholtz equation & Beam solution
- Geometrical optics
  - short wavelengths
  - Optical design dimensions larger than wavelength

SUMMARY LIGHT PROPAGATION

- Full wave electromagnetic field
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  - **Geometrical Optics**: spherical lenses
  - **Paraxial Helmholtz equation** & **Beam solution**
- Geometrical optics
  - short wavelengths
  - Optical design dimensions larger than wavelength

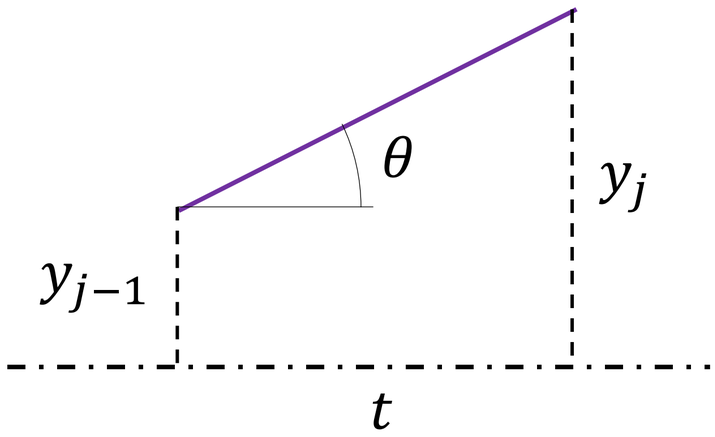
ABCD Matrices

Propagation of light through “thick” lenses BUT paraxial approx.

RAY TRACING

Conditions

- Circularly symmetric around optical axis.
- Ray are **Paraxial** & **Meridional**



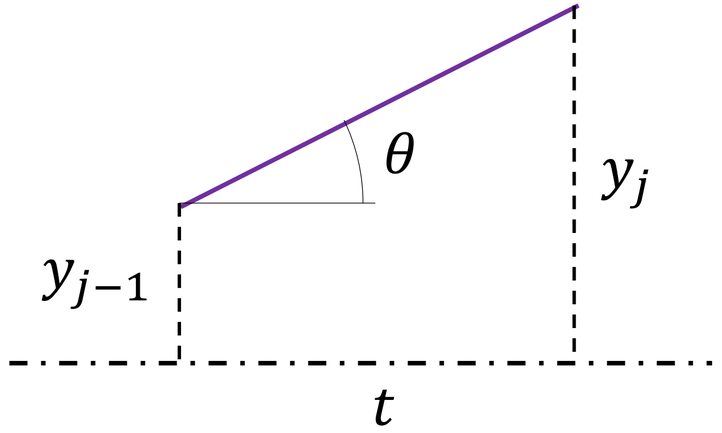
Rays are defined by:

- Angle θ
- Height from optical axis y

$$y_j = y_{j-1} + \theta\, t$$

RAY TRACING FOR SINGLE SURFACE

- Free space: straight propagation
- At surfaces: change direction

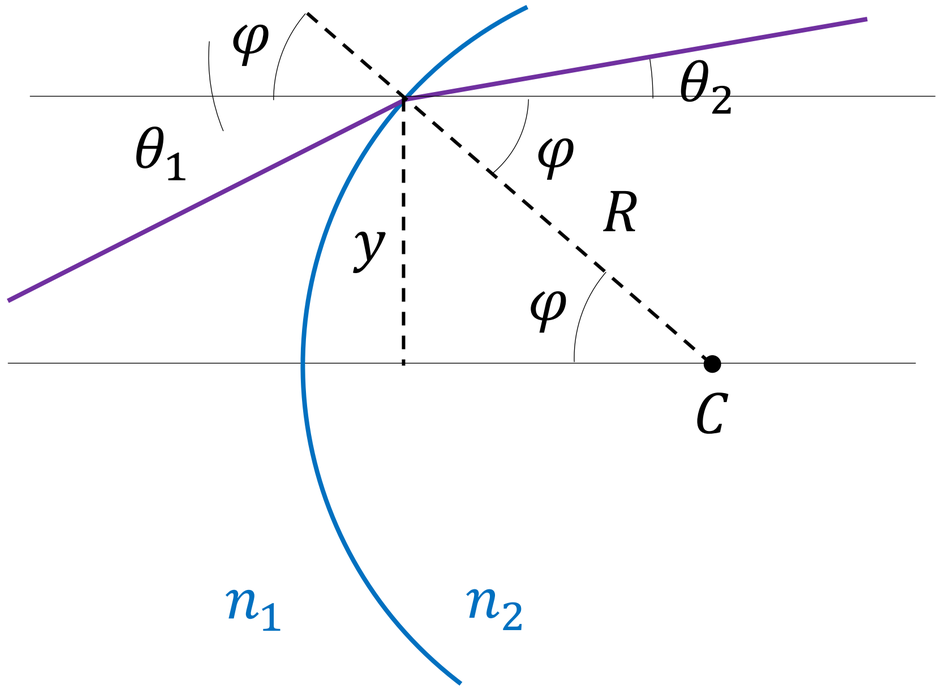


Snell's law:

$$n_1 \sin(\varphi + \theta_1) = n_2 \sin(\varphi + \theta_2)$$

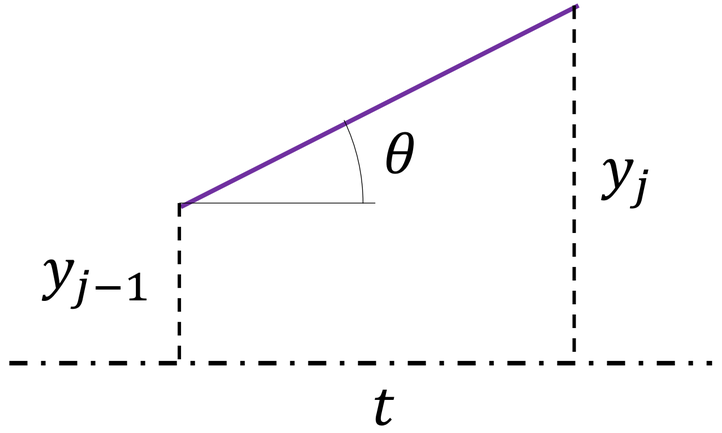
Paraxial approx: $\sin(\alpha) \approx \alpha$

$$n_1(\varphi + \theta_1) = n_2(\varphi + \theta_2)$$



RAY TRACING: THE ABCD-MATRIX

- Free space: straight propagation
- At surfaces: change direction

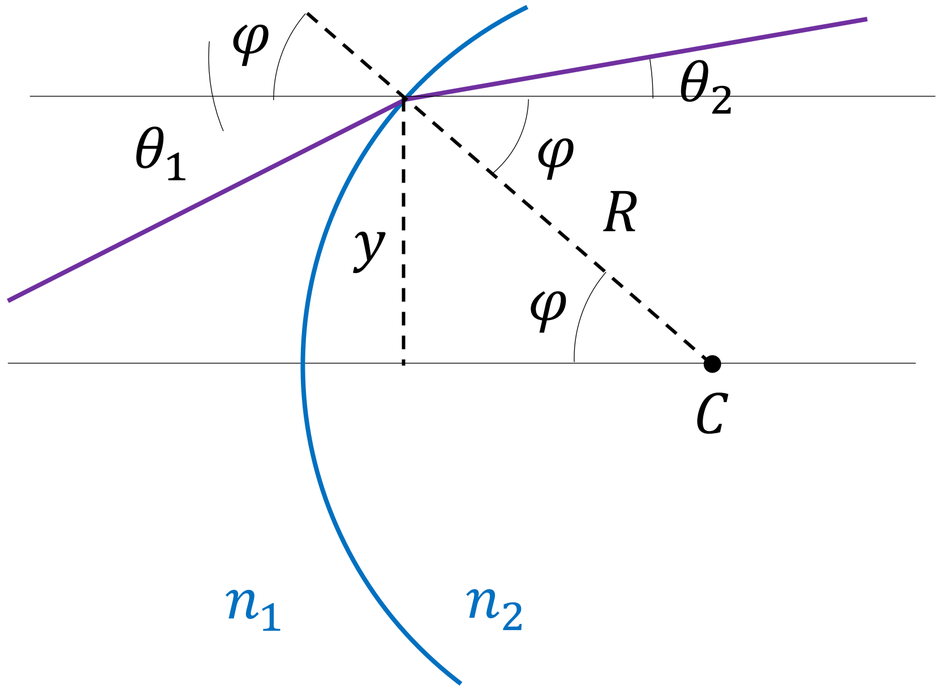


- **Relation ray heights and angles**

$$\begin{cases} y_2 = A\, y_1 + B\, \theta_1 \\ \theta_2 = C\, y_1 + D\, \theta_1 \end{cases}$$

OR

$$\begin{bmatrix} y_2 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} y_1 \\ \theta_1 \end{bmatrix}$$



ABCD-MATRIX: FREE SPACE

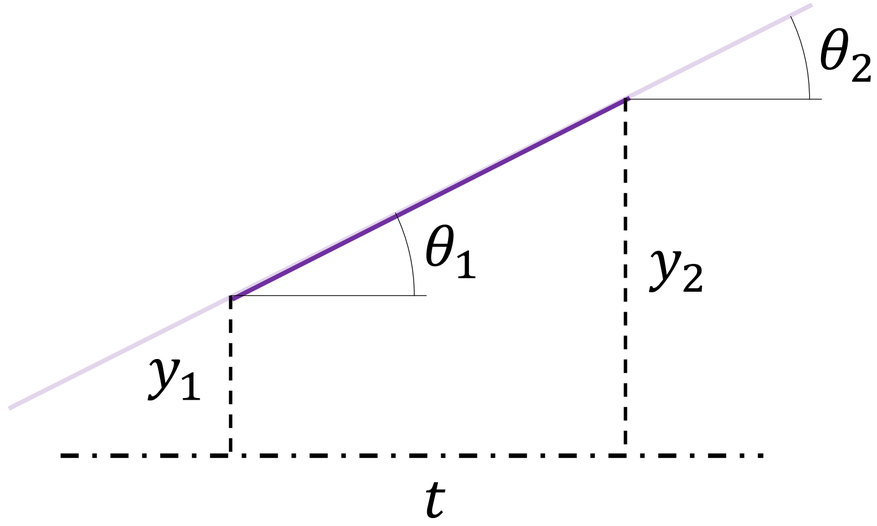
What are A, B, C, D?

$$\begin{bmatrix} y_2 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} y_1 \\ \theta_1 \end{bmatrix}$$

In free space:

$$\begin{cases} y_2 = y_1 + t\theta_1 \\ \theta_2 = \theta_1 \end{cases}$$

$$\Rightarrow \quad \begin{bmatrix} y_2 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ \theta_1 \end{bmatrix}$$



ABCD-MATRIX: PLANAR SURFACE

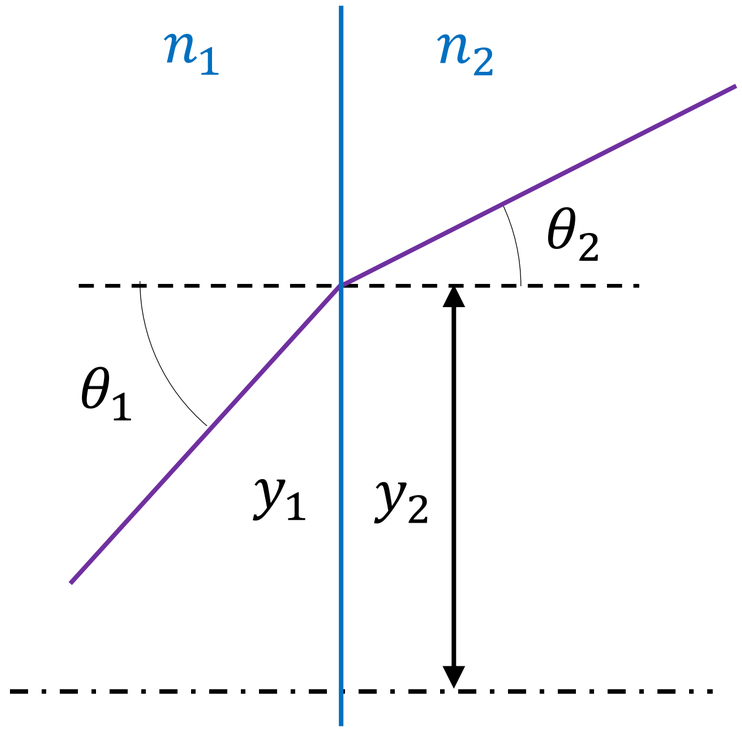
What are A, B, C, D?

$$\begin{bmatrix} y_2 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} y_1 \\ \theta_1 \end{bmatrix}$$

Snell's law (**paraxial** approx.)

$$n_2\theta_2 = n_1\theta_1$$

$$\begin{cases} y_2 = y_1 \\ \theta_2 = \dfrac{n_1}{n_2}\theta_1 \end{cases}$$



$$\Rightarrow \quad \begin{bmatrix} y_2 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \dfrac{n_1}{n_2} \end{bmatrix} \begin{bmatrix} y_1 \\ \theta_1 \end{bmatrix}$$

ABCD-MATRIX: SPHERICAL SURFACE

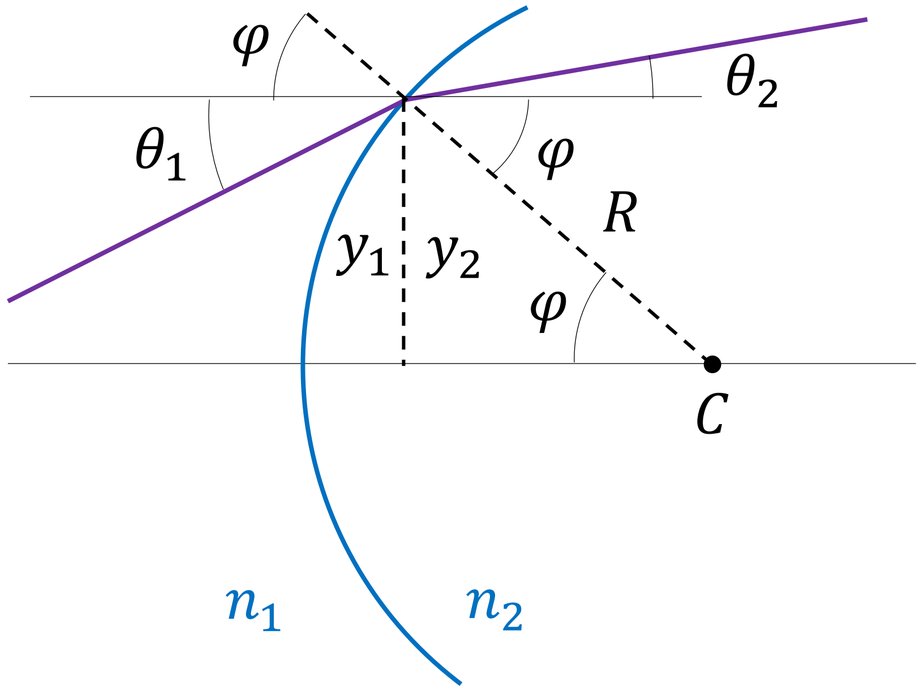
What are A, B, C, D?

$$\begin{bmatrix} y_2 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} y_1 \\ \theta_1 \end{bmatrix}$$

Snell's law & **paraxial** approx.

$$n_1(\varphi + \theta_1) = n_2(\varphi + \theta_2)$$

$$\varphi \approx \frac{y_1}{R}$$

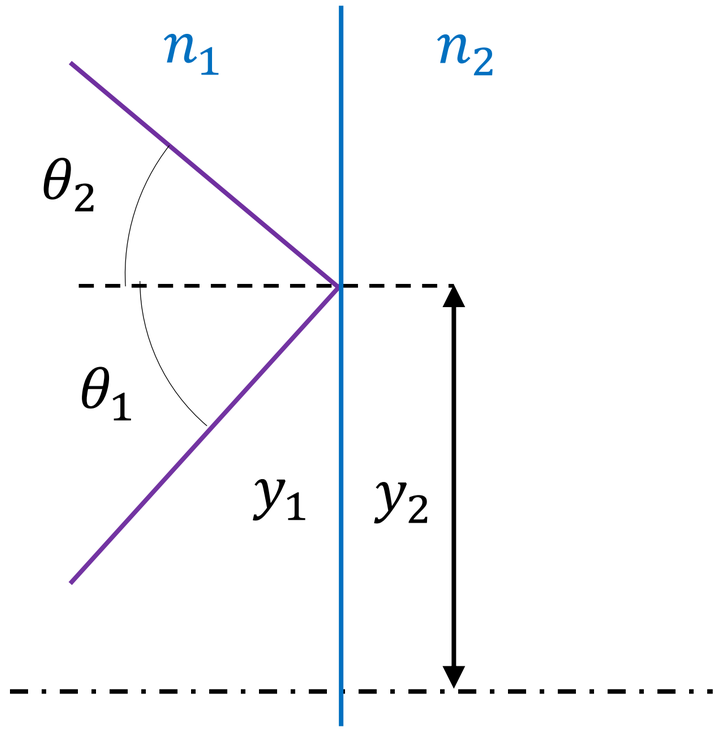


$$\Rightarrow \quad \theta_2 = \frac{-(n_2 - n_1)}{n_2}\frac{y_1}{R} + \frac{n_1}{n_2}\theta_1 \quad \Rightarrow \quad \begin{bmatrix} y_2 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{-(n_2 - n_1)}{n_2 R} & \frac{n_1}{n_2} \end{bmatrix} \begin{bmatrix} y_1 \\ \theta_1 \end{bmatrix}$$

ABCD-MATRIX: REFLECTION ON A PLANAR SURFACE

What are A, B, C, D?

$$\begin{bmatrix} y_2 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} y_1 \\ \theta_1 \end{bmatrix}$$



Angle changes sign $\theta_2 = -\theta_1$?

but **z-axis also flips**

$$\begin{cases} y_2 = y_1 \\ \theta_2 = \theta_1 \end{cases}$$

$$\Rightarrow \quad \begin{bmatrix} y_2 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ \theta_1 \end{bmatrix}$$

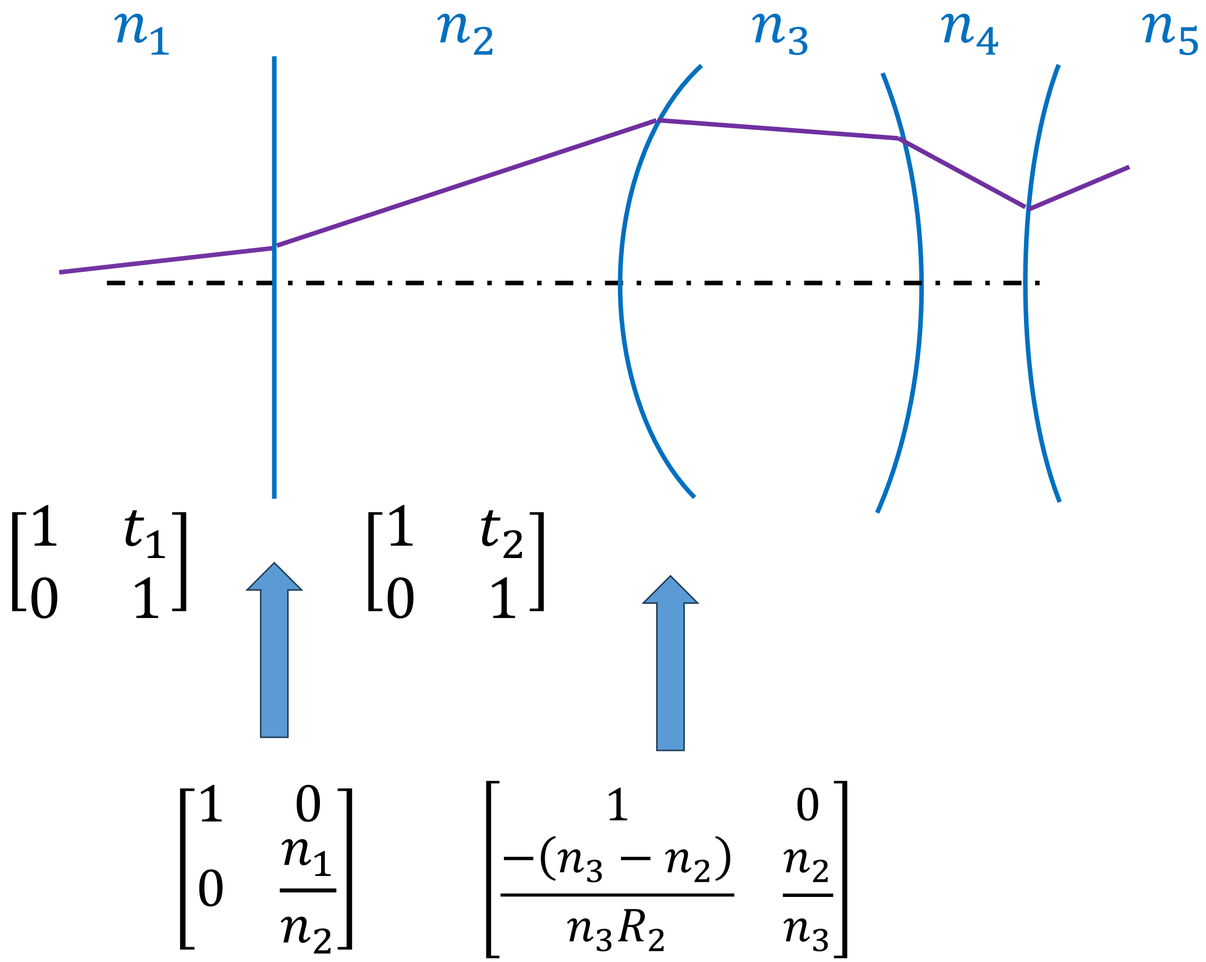
ABCD-MATRICES: PROPAGATION

ABCD-matrix for single surface:

$$\begin{bmatrix} y_2 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} y_1 \\ \theta_1 \end{bmatrix} = M_1 \begin{bmatrix} y_1 \\ \theta_1 \end{bmatrix}$$

System with multiple surfaces:

$$\begin{bmatrix} y_{N+1} \\ \theta_{N+1} \end{bmatrix} = M_N \dots M_2 M_1 \begin{bmatrix} y_1 \\ \theta_1 \end{bmatrix}$$



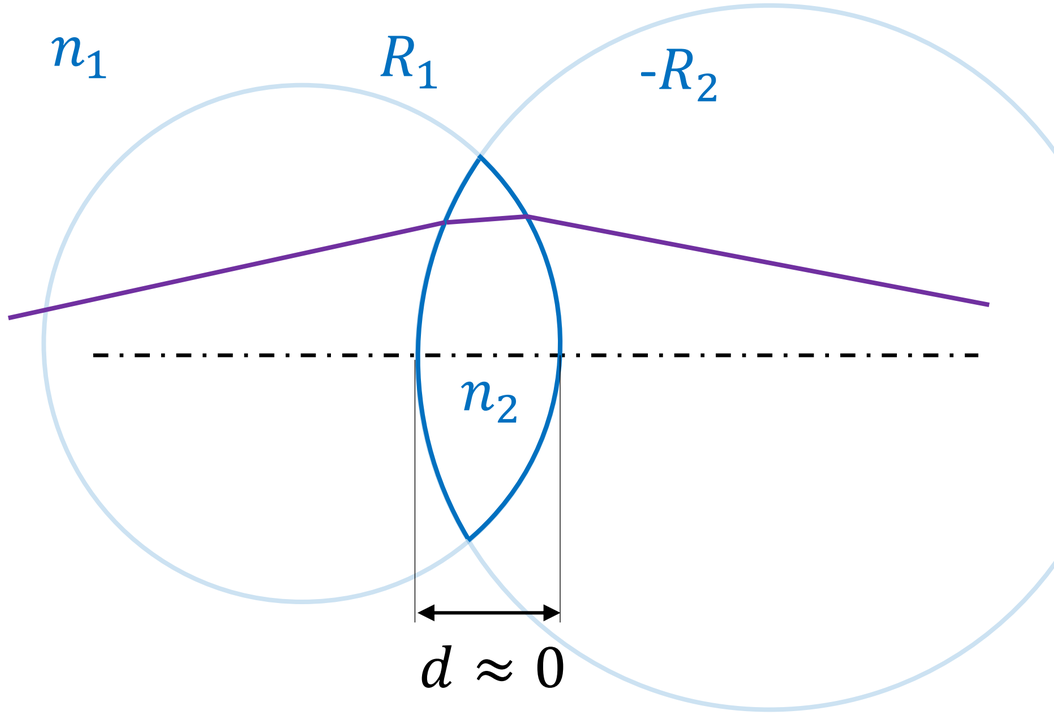
ABCD-MATRICES: THIN LENS

Two curved surfaces:

$$M = \begin{bmatrix} 1 & 0 \\ \frac{-(n_1 - n_2)}{n_1 R_2} & \frac{n_2}{n_1} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \frac{-(n_2 - n_1)}{n_2 R_1} & \frac{n_1}{n_2} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ \frac{-(n_2 - n_1)}{n_1}\left(\frac{1}{R_1} - \frac{1}{R_2}\right) & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{bmatrix}$$



Lensmaker's formula

$$\frac{1}{f} = \frac{(n_2 - n_1)}{n_1}\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

ABCD-MATRICES: THIN LENS IMAGING

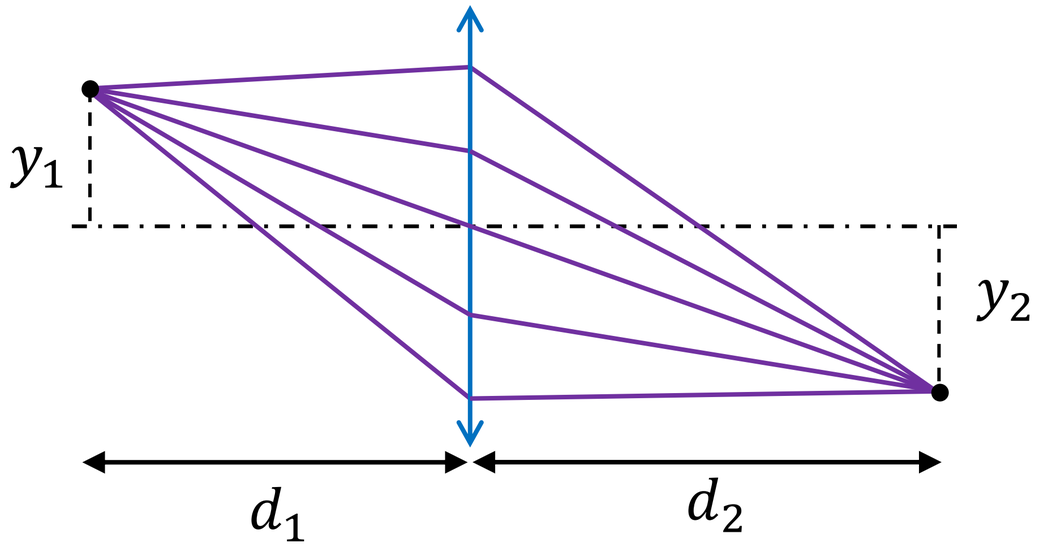
Free space 2 Free space 1

Thin lens

$$M = \begin{bmatrix} 1 & d_2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{bmatrix} \begin{bmatrix} 1 & d_1 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 - \frac{d_2}{f} & d_1 + d_2 - \frac{d_1 d_2}{f} \\ -\frac{1}{f} & 1 - \frac{d_1}{f} \end{bmatrix}$$

$$= \begin{bmatrix} 1 - \frac{d_2}{f} & 0 \\ -\frac{1}{f} & 1 - \frac{d_1}{f} \end{bmatrix}$$



Thin lens condition

$$\frac{1}{f} = \frac{1}{d_1} + \frac{1}{d_2}$$

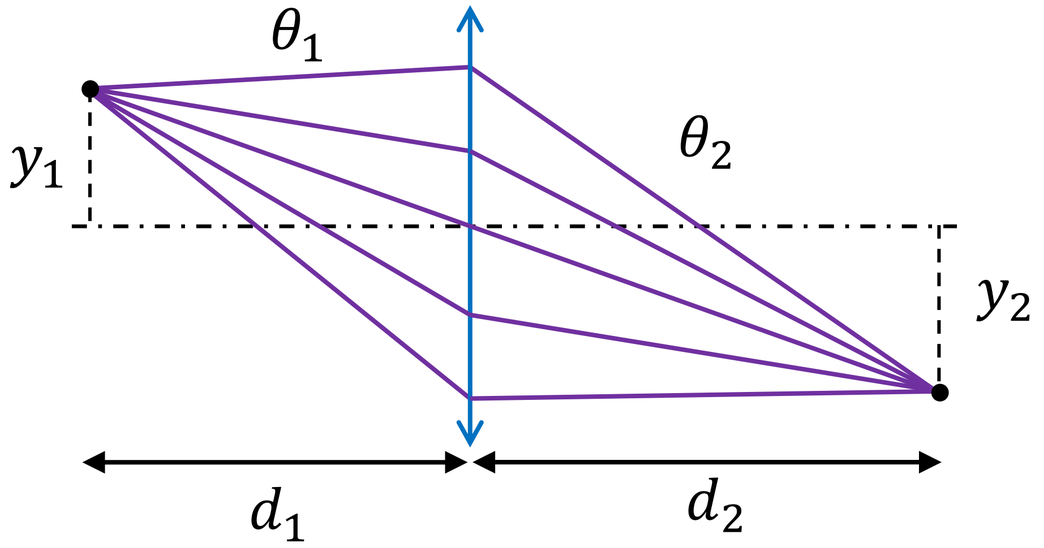
ABCD-MATRICES: THIN LENS IMAGING

Relation ray angles & heigths:

$$\begin{bmatrix} y_2 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 1 - \frac{d_2}{f} & 0 \\ -\frac{1}{f} & 1 - \frac{d_1}{f} \end{bmatrix} \begin{bmatrix} y_1 \\ \theta_1 \end{bmatrix}$$

Ray height independent of angle

$$y_2 = y_1 \left(1 - \frac{d_2}{f}\right)$$



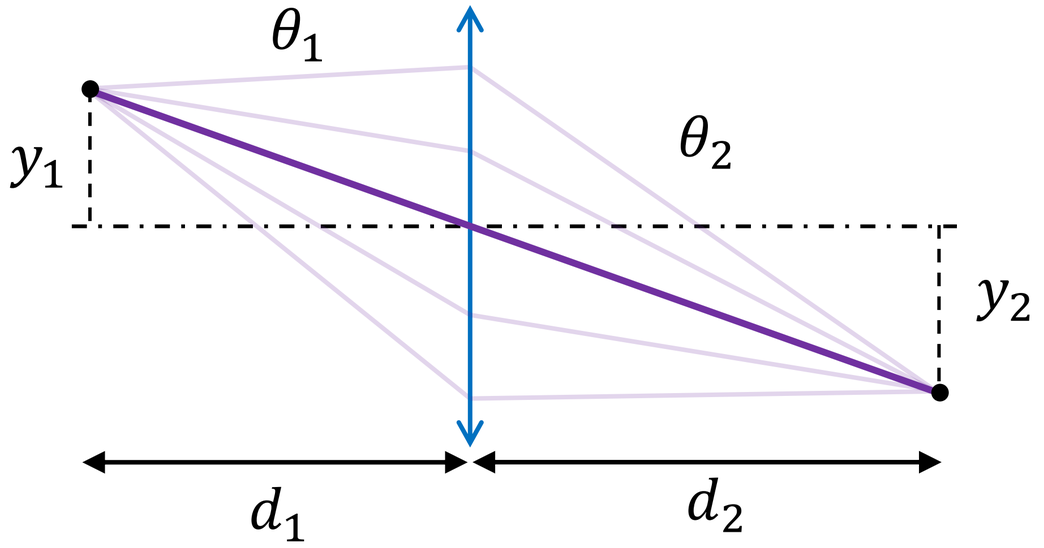
ABCD-MATRICES: THIN LENS IMAGING

Relation ray angles & heigths:

$$\begin{bmatrix} y_2 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 1 - \frac{d_2}{f} & 0 \\ -\frac{1}{f} & 1 - \frac{d_1}{f} \end{bmatrix} \begin{bmatrix} y_1 \\ \theta_1 \end{bmatrix}$$

Ray height independent of angle

$$y_2 = y_1 \left(1 - \frac{d_2}{f}\right)$$



Ray through center: relation between heights and distances

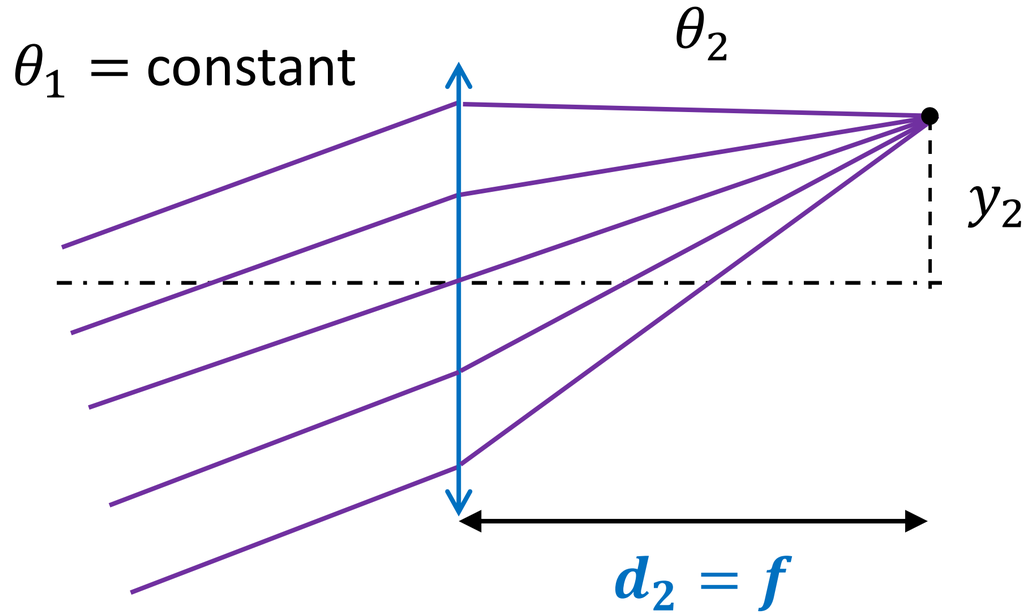
$$\frac{y_2}{d_2} = \frac{y_1}{d_1}$$

ABCD-MATRICES: THIN LENS IMAGING

Relation ray angles & heights:

$$\begin{bmatrix} y_2 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 1 - \frac{d_2}{f} & d_1 + d_2 - \frac{d_1 d_2}{f} \\ -\frac{1}{f} & 1 - \frac{d_1}{f} \end{bmatrix} \begin{bmatrix} y_1 \\ \theta_1 \end{bmatrix}$$

0 d_2



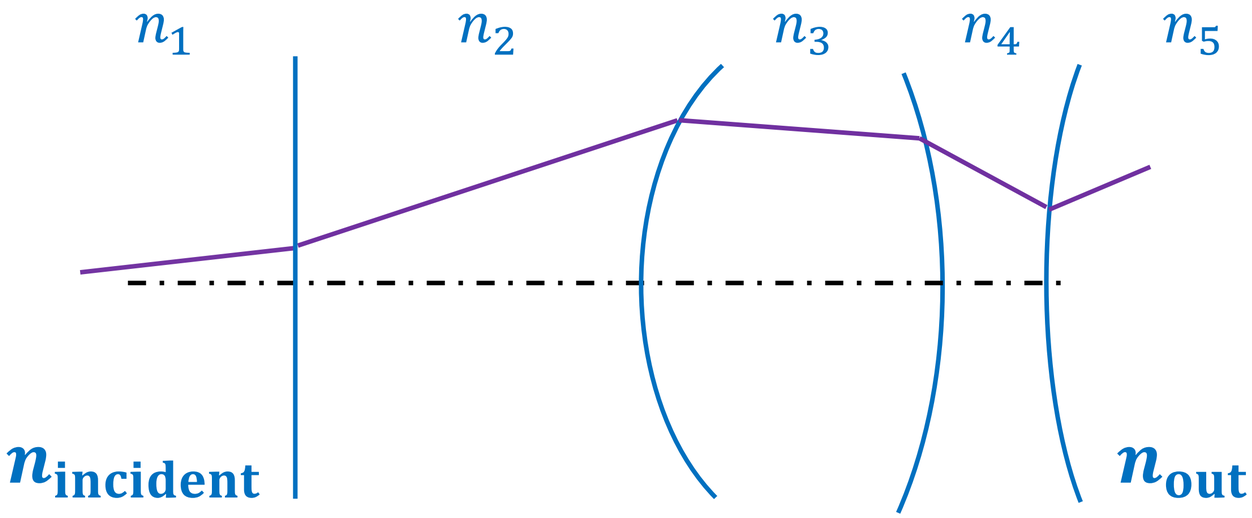
Ray height y_2 independent of height y_1

$$y_2 = B\ \theta_1 = d_2 \theta_1$$

THE LAGRANGE INVARIANT

System multiple surfaces

$$\begin{bmatrix} y_{N+1} \\ \theta_{N+1} \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} y_1 \\ \theta_1 \end{bmatrix}$$



ABCD-matrix:

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = M = M_N \ldots M_2 M_1$$

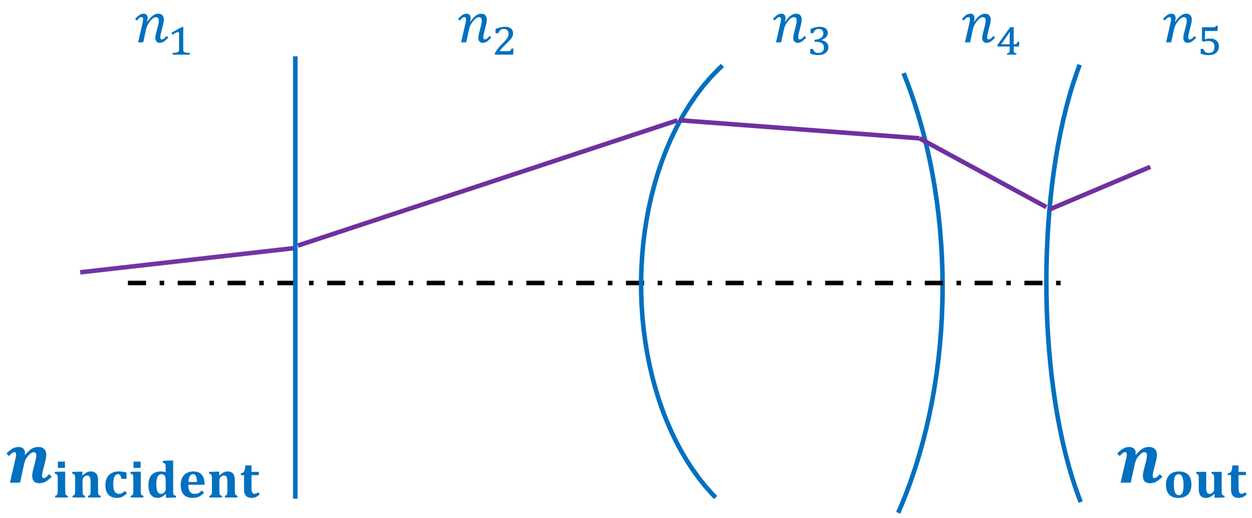
Constant:

$$\det(M) = \frac{n_{\text{incident}}}{n_{\text{out}}}$$

THE LAGRANGE INVARIANT

System multiple surfaces

$$\begin{bmatrix} y_{N+1} \\ \theta_{N+1} \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} y_1 \\ \theta_1 \end{bmatrix}$$



ABCD-matrix:

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = M = M_N \dots M_2 M_1 \qquad \det(M) = \det(M_N) \cdot \dots \cdot \det(M_2) \cdot \det(M_1)$$

Constant:

$$\det(M) = \frac{n_{\text{incident}}}{n_{\text{out}}}$$

THE LAGRANGE INVARIANT

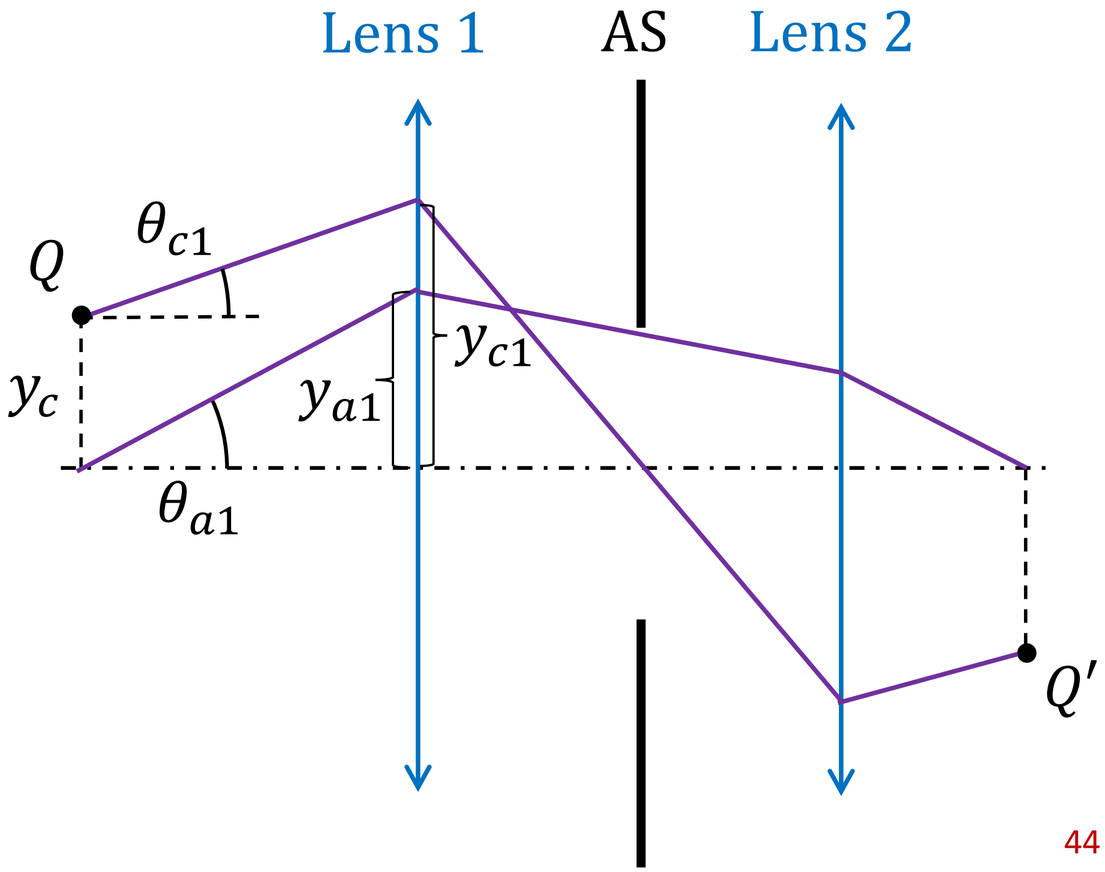
Langrange invariant: $\mathrm{LI} = y_{c1}\theta_{a1} - y_{a1}\theta_{c1}$ is a constant

Two independent rays:

- Chief ray (y_{c1}, θ_{c1})
- Axial ray (y_{a1}, θ_{a1})

Both rays obey the ABCD-matrix

$$\begin{bmatrix} y_{N+1} \\ \theta_{N+1} \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} y_1 \\ \theta_1 \end{bmatrix}$$

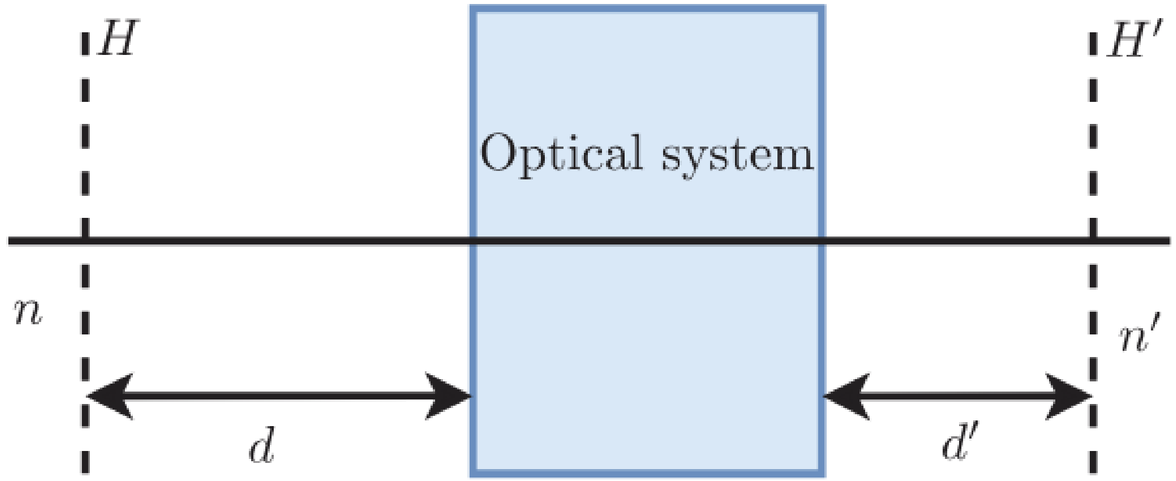


OPTICAL SYSTEM AND PRINCIPAL PLANES

- Matrix calculations less insight

- Simplifying ABCD-matrix

$$\begin{bmatrix} y' \\ \theta' \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} y \\ \theta \end{bmatrix}$$

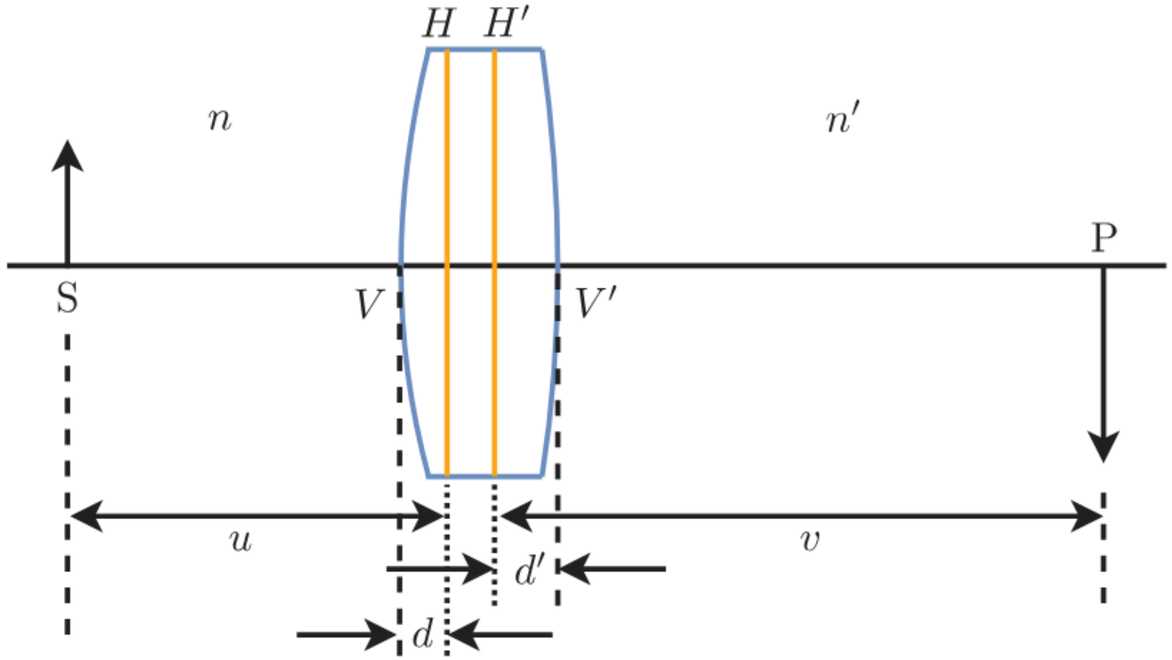


$$M_{HH'} = \begin{bmatrix} 1 & d' \\ 0 & 1 \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} 1 & d \\ 0 & 1 \end{bmatrix}$$

OPTICAL SYSTEM AND PRINCIPAL PLANES

- Matrix calculations less insight
- Simplifying ABCD-matrix

$$\begin{bmatrix} y' \\ \theta' \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} y \\ \theta \end{bmatrix}$$

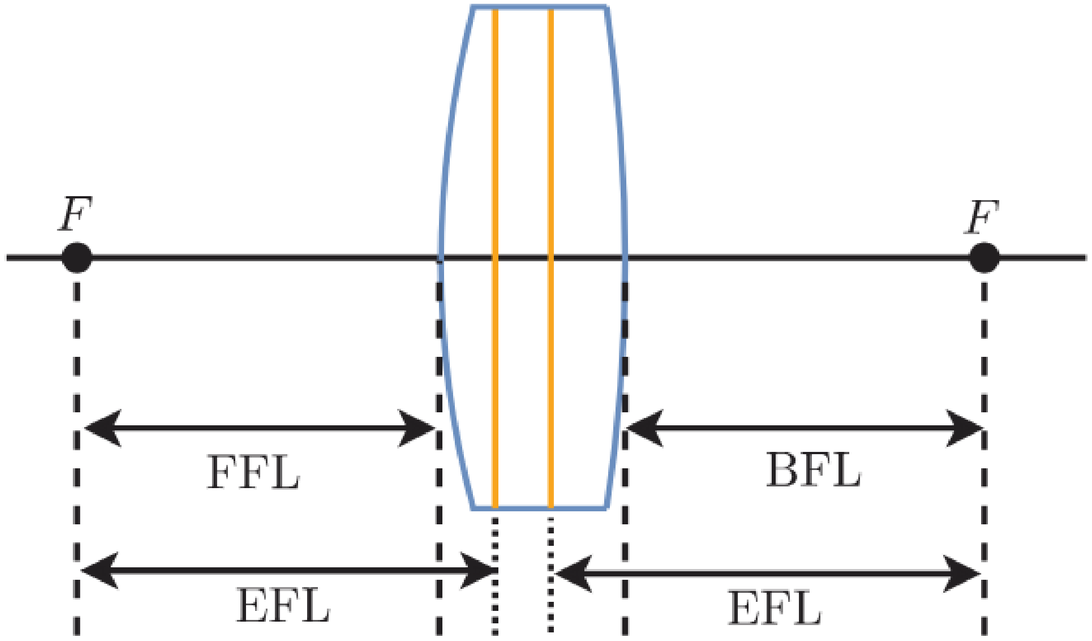


$$\begin{bmatrix} A & 0 \\ C & D \end{bmatrix} = \begin{bmatrix} 1 & v \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -\frac{P}{n'} & \frac{n}{n'} \end{bmatrix} \begin{bmatrix} 1 & u \\ 0 & 1 \end{bmatrix}$$

PRINCIPAL PLANES

ABCD-matrix

$$\begin{bmatrix} y' \\ \theta' \end{bmatrix} = \begin{bmatrix} 1 - \frac{vP}{n'} & u - \frac{uvP}{n'} + \frac{vn}{n'} \\ -\frac{P}{n'} & -\frac{uP}{n'} + \frac{n}{n'} \end{bmatrix} \begin{bmatrix} y \\ \theta \end{bmatrix}$$



- Imaging system:

$$B = u - \frac{uvP}{n'} + \frac{vn}{n'} = 0 \Rightarrow \boldsymbol{\frac{n}{u} + \frac{n'}{v} = P}$$

- BFL: Back Focal Length
- FFL: Front Focal Length
- EFL: Effective Focal Length

Practical Calculations

Matrix multiplications, determinants, ABCD matrices, Principal Planes